NEET

Time : 3 H 20 min
MM : 720

Solved Paper 2023

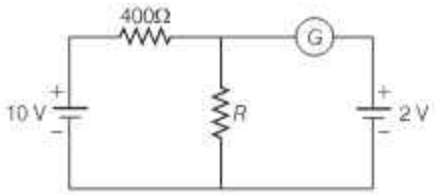
Instructions

- This question paper contains 200 Multiple Choice Question from the subjects **Physics, Chemistry** & **Biology** (Botany & Zoology). There are 50 questions in each subject, divided into two Sections : Section-A and Section B.
- **Section A** contains 35 questions in each subject (Q. Nos. 1 to 35, 51 to 85, 101 to 135 and 151 to 185) and all questions are compulsory.
- **Section B** contains 15 questions from each subject (Q. Nos. 36 to 50, 86 to 100, 136 to 150 and 186 to 200) out of which 10 questions are to be attempted.
- Each question carries 4 Marks. For each correct response, the candidate will get 4 marks. For each incorrect response 1 mark will be deducted from the total scores.

PHYSICS

Section-A

1. If the galvanometer (G) does not show any deflection in the circuit shown, the value of R is given by



(a) 50 Ω (b) 100 Ω (c) 400 Ω (d) 200 Ω

2. A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the AC ripple from the rectified output?
(a) p-n junction diodes
(b) Capacitor
(c) Load resistance
(d) A centre-tapped transformer

3. A Carnot engine has an efficiency of 50% when its source is at a temperature 327°C. The temperature of the sink is
(a) 15°C (b) 100°C (c) 200°C (d) 27°C

4. Light travels a distance x in time t_1 in air and $10x$ in time t_2 in another denser medium. What is the critical angle for this medium?
(a) $\sin^{-1}\left(\frac{10t_2}{t_1}\right)$ (b) $\sin^{-1}\left(\frac{t_1}{10t_2}\right)$
(c) $\sin^{-1}\left(\frac{10t_1}{t_2}\right)$ (d) $\sin^{-1}\left(\frac{t_2}{t_1}\right)$

5. The angular acceleration of a body, moving along the circumference of a circle, is
(a) along the radius towards the centre
(b) along the tangent to its position
(c) along the axis of rotation
(d) along the radius, away from centre

6. The venturimeter works on
(a) Bernoulli's principle
(b) the principle of parallel axes
(c) the principle of perpendicular axes
(d) Huygen's principle

***7.** The half-life of a radioactive substance is 20 minutes. In how much time, the activity of substance drops to (1/16)th of its initial value?
(a) 40 minutes (b) 60 minutes
(c) 80 minutes (d) 20 minutes

8. The errors in the measurement which arise due to unpredictable fluctuations in temperature and voltage supply are
(a) personal errors (b) least count errors
(c) random error (d) instrumental errors

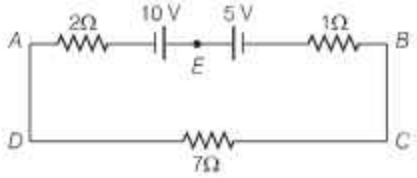
9. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 NC^{-1}. It experiences a torque equal to 4 N m. Calculate the magnitude of charge on the dipole, if the dipole length is 2 cm.
(a) 6 mC (b) 4 mC (c) 2 mC (d) 8 mC

10. A football player is moving southward and suddenly turns eastward with the same speed to avoid an opponent. The force that acts on the player while turning is
(a) along northward (b) along north-east
(c) along south-west (d) along eastward

Note *(*) Marked questions are from the topics not included in latest NEET syllabus.*

11. In hydrogen spectrum, the shortest wavelength in the Balmer series is λ. The shortest wavelength in the Bracket series is
(a) 4λ (b) 9λ (c) 16λ (d) 2λ

12. The magnitude and direction of the current in the following circuit is



(a) 0.5 A from A to B through E
(b) $\frac{5}{9}$ A from A to B through E
(c) 1.5 A from B to A through E
(d) 0.2 A from B to A through E

13. A 12 V, 60 W lamp is connected to the secondary of a step down transformer, whose primary is connected to AC mains of 220 V. Assuming the transformer to be ideal, what is the current in the primary winding?
(a) 2.7 A (b) 3.7 A (c) 0.37 A (d) 0.27 A

14. The work functions of caesium (Cs), Potassium (K) and sodium (Na) are 2.14 eV, 2.30 eV and 2.75 eV respectively. If incident electromagnetic radiation has an incident energy of 2.20 eV. Which of these photosensitive surfaces may emit photoelectrons?
(a) Both Na and K (b) K only
(c) Na only (d) Cs only

15. The potential energy of a long spring when stretched by 2 cm is U. If the spring is stretched by 8 cm, potential energy stored in it will be
(a) $4U$ (b) $8U$ (c) $16U$ (d) $2U$

16. The net magnetic flux through any closed surface is
(a) positive (b) infinity (c) negative (d) zero

17. The amount of energy required to form a soap bubble of radius 2 cm from a soap solution is nearly (surface tension of soap solution $= 0.03\ \text{N m}^{-1}$)
(a) 5.06×10^{-4} J (b) 3.01×10^{-4} J
(c) 50.1×10^{-4} J (d) 30.16×10^{-4} J

18. Let a wire be suspended from the ceiling (rigid support) and stretched by a weight w attached at its free end. The longitudinal stress at any point of cross-sectional area A of the wire is
(a) w/A (b) $w/2A$ (c) zero (d) $2w/A$

19. Two bodies of mass m and $9m$ are placed at a distance R. The gravitational potential on the line joining the bodies where the gravitational field equals zero, will be (G = gravitational constant)
(a) $-\frac{12Gm}{R}$ (b) $-\frac{16Gm}{R}$ (c) $-\frac{20Gm}{R}$ (d) $-\frac{8Gm}{R}$

20. The temperature of a gas is –50°C. To what temperature the gas should be heated so that the rms speed is increased by 3 times?
(a) 3295°C (b) 3097 K (c) 223 K (d) 669°C

21. Given below are two statements.
Statement I Photovoltaic devices can convert optical radiation into electricity.
Statement II Zener diode is designed to operate under reverse bias in breakdown region.
In the light of the above statements, choose the ***most appropriate*** answer from the options given below
(a) Both Statement I and Statement II are incorrect.
(b) Statement I is correct but Statement II is incorrect.
(c) Statement I is incorrect but Statement II is correct.
(d) Both Statement I and Statement II are correct.

22. If $\oint \mathbf{E} \cdot \mathbf{dS} = 0$ over a surface, then:
(a) the magnitude of electric field on the surface is constant.
(b) all the charges must necessarily be inside the surface.
(c) the electric field inside the surface is necessarily uniform.
(d) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

23. In a series L-C-R circuit, the inductance L is 10 mH, capacitance C is 1 μF and resistance R is 100 Ω. The frequency at which resonance occurs is
(a) 15.9 kHz (b) 1.59 rad/s (c) 1.59 kHz (d) 15.9 rad/s

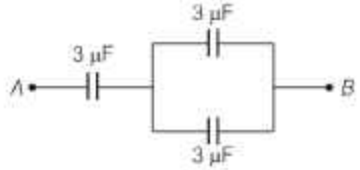
24. An AC source is connected to a capacitor C. Due to decrease in its operating frequency
(a) displacement current increases.
(b) displacement current decreases.
(c) capacitive reactance remains constant.
(d) capacitive reactance decreases.

25. A bullet is fired from a gun at the speed of 280 m s^{-1} in the direction 30° above the horizontal.
The maximum height attained by the bullet is
($g = 9.8\ \text{m s}^{-2}$, $\sin 30^\circ = 0.5$)
(a) 2000 m (b) 1000 m (c) 3000 m (d) 2800 m

26. A vehicle travels half the distance with speed v and the remaining distance with speed $2v$. Its average speed is
(a) $\frac{2v}{3}$ (b) $\frac{4v}{3}$ (c) $\frac{3v}{4}$ (d) $\frac{v}{3}$

***27.** Resistance of a carbon resistor determined from colour codes is $(22000 \pm 5\%)\ \Omega$. The colour of third band must be
(a) green (b) orange (c) yellow (d) red

28. The equivalent capacitance of the system shown in the following circuit is



(a) 3 μF (b) 6 μF (c) 9 μF (d) 2 μF

29. The minimum wavelength of X-rays produced by an electron accelerated through a potential difference of V volts is proportional to

(a) $\frac{1}{V}$ (b) $\frac{1}{\sqrt{V}}$ (c) V^2 (d) $\sqrt{V}$

30. The magnetic energy stored in an inductor of inductance 4 μH carrying a current of 2 A is

(a) 4 mJ (b) 8 mJ (c) 8 μJ (d) 4 μJ

31. For Young's double slit experiment, two statements are given below:

Statement I If screen is moved away from the plane of slits, angular separation of the fringes remains constant.

Statement II If the monochromatic source is replaced by another monochromatic source of larger wavelength, the angular separation of fringes decreases.

In the light of the above statements, choose the correct answer from the options given below

(a) Both Statement I and Statement II are false.
(b) Statement I is true but Statement II is false.
(c) Statement I is false but Statement II is true.
(d) Both Statement I and Statement II are true.

32. The ratio of radius of gyration of a solid sphere of mass M and radius R about its own axis to the radius of gyration of the thin hollow sphere of same mass and radius about its axis is

(a) 5 : 3 (b) 2 : 5 (c) 5 : 2 (d) 3 : 5

33. In a plane electromagnetic wave travelling in free space, the electric field component oscillates sinusoidally at a frequency of 2.0×10^{10} Hz and amplitude 48 V m^{-1}. Then the amplitude of oscillating magnetic field is (Speed of light in free space $= 3 \times 10^8$ m s^{-1})

(a) 1.6×10^{-8} T (b) 1.6×10^{-7} T
(c) 1.6×10^{-6} T (d) 1.6×10^{-9} T

34. A metal wire has mass (0.4 ± 0.002) g, radius (0.3 ± 0.001) mm and length (5 ± 0.02) cm. The maximum possible percentage error in the measurement of density will nearly be

(a) 1.3% (b) 1.6% (c) 1.4% (d) 1.2%

35. The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is

(a) 2 : 1 (b) 1 : 3 (c) 3 : 1 (d) 1 : 2

Section-B

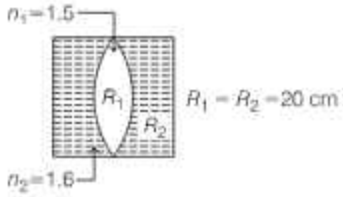
36. Two thin lenses are of same focal lengths f, but one is convex and the other one is concave. When they are placed in contact with each other, the equivalent focal length of the combination will be

(a) $f/4$ (b) $f/2$ (c) infinite (d) Zero

37. A satellite is orbiting just above the surface of the earth with period T. If d is the density of the earth and G is the universal constant of gravitation, the quantity $\frac{3\pi}{Gd}$ represents

(a) T^2 (b) T^3 (c) $\sqrt{T}$ (d) T

38. In the figure shown here, what is the equivalent focal length of the combination of lenses (Assume that, all layers are thin)?



(a) – 40 cm (b) – 100 cm (c) – 50 cm (d) 40 cm

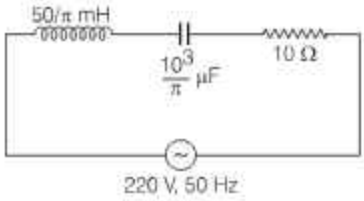
39. A bullet from a gun is fired on a rectangular wooden block with velocity u. When bullet travels 24 cm through the block along its length horizontally, velocity of bullet becomes $\frac{u}{3}$. Then it further penetrates into the block in the same direction before coming to rest exactly at the other end of the block. The total length of the block is

(a) 24 cm (b) 28 cm (c) 30 cm (d) 27 cm

40. 10 resistors, each of resistance R are connected in series to a battery of emf E and negligible internal resistance. Then those are connected in parallel to the same battery, the current is increased n times. The value of n is

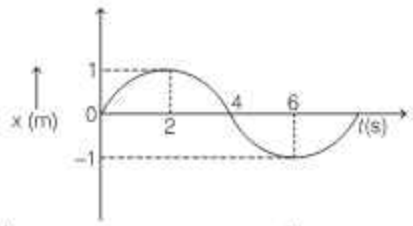
(a) 100 (b) 1 (c) 1000 (d) 10

41. The net impedance of circuit (as shown in figure) will be



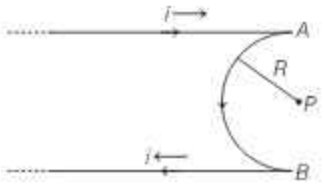
(a) 15 Ω (b) $5\sqrt{5}$ Ω (c) 25 Ω (d) $10\sqrt{2}$ Ω

42. The x-t graph of a particle performing simple harmonic motion is shown in the figure. The acceleration of the particle at $t = 2$ s is



(a) $-\frac{\pi^2}{8}$ m s^{-2} (b) $\frac{\pi^2}{16}$ m s^{-2}
(c) $-\frac{\pi^2}{16}$ m s^{-2} (d) $\frac{\pi^2}{8}$ m s^{-2}

43. A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by



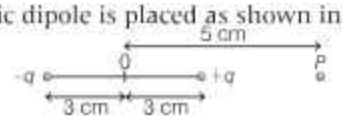
(a) $\frac{\mu_0 i}{4R}$ pointed away from the page

(b) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed away from page

(c) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page

(d) $\frac{\mu_0 i}{4R}$ pointed into the page

44. An electric dipole is placed as shown in the figure.



The electric potential (in 10^2 V) at point P due to the dipole is (ε_0 = permittivity of free space and $\frac{1}{4\pi\varepsilon_0} = K$)

(a) $\left(\frac{5}{8}\right)qk$ (b) $\left(\frac{8}{5}\right)qk$

(c) $\left(\frac{8}{3}\right)qk$ (d) $\left(\frac{3}{8}\right)qk$

45. The resistance of platinum wire at 0°C is 2 Ω and 6.8 Ω at 80°C. The temperature coefficient of resistance of the wire is

(a) $3\times10^{-3}\,°C^{-1}$ (b) $3\times10^{-2}\,°C^{-1}$

(c) $3\times10^{-1}\,°C^{-1}$ (d) $3\times10^{-4}\,°C^{-1}$

46. A wire carrying a current I along the positive X-axis has length L. It is kept in a magnetic field $\mathbf{B} = (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 4\hat{\mathbf{k}})$ T. The magnitude of the magnetic force acting on the wire is

(a) $\sqrt{5}\,IL$ (b) $5\,IL$ (c) $\sqrt{3}\,IL$ (d) $3\,IL$

47. Calculate the maximum acceleration of a moving car so that a body lying on the floor of the car remains stationary. The coefficient of static friction between the body and the floor is 0.15 ($g = 10\ m\,s^{-2}$).

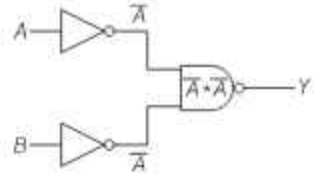
(a) $150\ m\,s^{-2}$ (b) $1.5\ m\,s^{-2}$

(c) $50\ m\,s^{-2}$ (d) $1.2\ m\,s^{-2}$

48. The radius of inner most orbit of hydrogen atom is 5.3×10^{-11} m. What is the radius of third allowed orbit of hydrogen atom?

(a) 1.06 Å (b) 1.59 Å (c) 4.77 Å (d) 0.53 Å

49. For the following logic circuit, the truth table is



(a)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(b)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

(c)

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(d)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

50. A horizontal bridge is built across a river. A student standing on the bridge throws a small ball vertically upwards with a velocity 4 $m\,s^{-1}$. The ball strikes the water surface after 4 s. The height of bridge above water surface is (Take, $g = 10\ m\,s^{-2}$)

(a) 60 m (b) 64 m (c) 68 m (d) 56 m

CHEMISTRY

Section-A

51. The relation between n_m (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (l), is

(a) $l = 2n_m + 1$ (b) $n_m = 2l^2 + l$

(c) $n_m = l + 2$ (d) $l = \frac{n_m - 1}{2}$

52. Amongst the given options which of the following molecules/ions acts as a Lewis acid?

(a) H_2O (b) BF_3 (c) OH^- (d) NH_3

***53.** Which of the following statements are not correct?

A. Hydrogen is used to reduce heavy metal oxides to metals.

B. Heavy water is used to study reaction mechanism.

C. Hydrogen is used to make saturated fats from oils.

D. The H—H bond dissociation enthalpy is lowest as compared to a single bond between two atoms of any element.

E. Hydrogen reduces oxides of metals that are more active than iron.

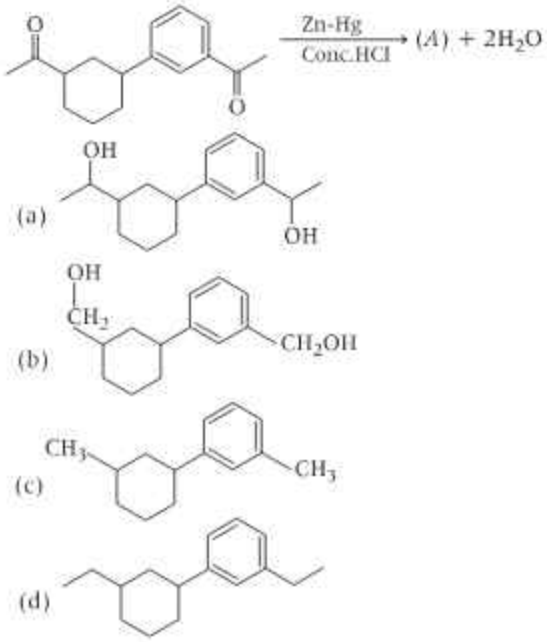
Choose the **most appropriate** answer from the options given below:

(a) B, D only (b) D, E only
(c) A, B, C only (d) B, C, D, E only

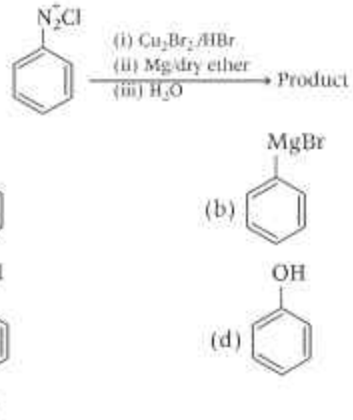
54. Which one of the following statements is correct?
(a) All enzymes that utilise ATP in phosphate transfer require Ca as the cofactor.
(b) The bone in human body is an inert and unchanging substance.
(c) Mg plays roles in neuromuscular function and interneuronal transmission.
(d) The daily requirement of Mg and Ca in the human body is estimated to be 0.2-0.3 g.

55. Homoleptic complex from the following complexes is
(a) diamminechloridonitrito-N-platinum (II)
(b) pentaamminecarbonatocobalt (III) chloride
(c) triamminetriaquachromium (III) chloride
(d) potassium trioxalatoaluminate (III)

56. Identify product (*A*) in the following reaction.



57. Identify the product in the following reaction.



58. The correct order of energies of molecular orbitals of N_2 molecule, is
(a) $\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$
(b) $\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \sigma^* 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y)$
(c) $\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma 2p_z < \sigma^* 2p_z$
(d) $\sigma 1s < \sigma^*1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

59. For a certain reaction, the rate $= k[A]^2[B]$, when the initial concentration of *A* is tripled keeping concentration of *B* constant, the initial rate would
(a) increase by a factor of six.
(b) increase by a factor of nine.
(c) increase by a factor of three.
(d) decrease by a factor of nine.

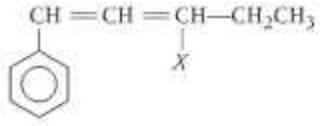
60. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) In equation $\Delta_r G = -nFE_{cell}$, the value of $\Delta_r G$ depends on *n*.

Reason (R) E_{cell} is an intensive property and $\Delta_r G$ is an extensive property.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true and R is not the correct explanation of A.
(b) A is true but R is false.
(c) A is false but R is true.
(d) Both A and R are true and R is the correct explanation of A.

61. The given compound

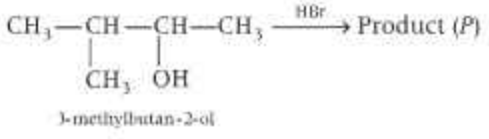


is an example of
(a) aryl halide (b) allylic halide
(c) vinylic halide (d) benylic halide

***62.** Which amongst the following molecules on polymerisation produces neoprene?



63. Consider the following reaction and identify the product (*P*).



(a) $CH_3CH=CH-CH_3$ (b) $CH_3-\underset{CH_3}{CH}-\underset{Br}{CH}-CH_3$

(c) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CH_2Br$ (d) $CH_3-\underset{CH_3}{\overset{Br}{C}}-CH_2-CH_3$

64. The conductivity of centimolar solution of KCl at 25°C is 0.0210 $ohm^{-1}cm^{-1}$ and the resistance of the cell containing the solution at 25°C is 60 ohm. The value of cell constant is

(a) 3.28 cm^{-1} (b) 1.26 cm^{-1} (c) 3.34 cm^{-1} (d) 1.34 cm^{-1}

65. Match List-I with List-II.

	List-I		List-II
A.	Coke	I.	Carbon atoms are sp^3-hybridised.
B.	Diamond	II.	Used as a dry lubricant.
C.	Fullerene	III.	Used as a reducing agent.
D.	Graphite	IV.	Cage like molecules.

Choose the correct answer from the options given below:

(a) A-IV, B-I, C-II, D-III (b) A-III, B-I, C-IV, D-II

(c) A-III, B-IV, C-I, D-II (d) A-II, B-IV, C-I, D-III

66. The element expected to form largest ion to achieve the nearest noble gas configuration is

(a) F (b) N (c) Na (d) O

67. Which of the following reactions will not give primary amine as the product?

(a) $CH_3CN \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4}$ Product

(b) $CH_3NC \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4}$ Product

(c) $CH_3CONH_2 \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4}$ Product

(d) $CH_3CONH_2 \xrightarrow{Br_2/KOH}$ Product

68. The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are

(a) 12, 3, 0 (b) 11, 3, 1 (c) 12, 2, 1 (d) 11, 2, 0

69. Given below are two statements.

Statement I A unit formed by the attachment of a base to 1'-position of sugar is known as nucleoside.

Statement II When nucleoside is linked to phosphoric acid at 5'-position of sugar moiety, we get nucleotide.

In the light of the above statements, choose the correct answer from the options given below:

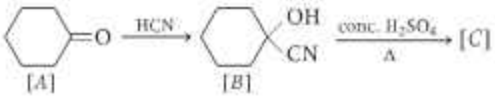
(a) Both Statement I and Statement II are false.

(b) Statement I is true but Statement II is false.

(c) Statement I is false but Statement II is true.

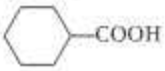
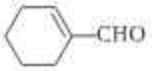
(d) Both Statement I and Statement II are true.

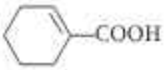
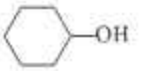
70. Complete the following reaction:



[A] [B]

is..... .

(a)  (b) 

(c)  (d) 

71. Taking stability as the factor, which one of the following represents correct relationship?

(a) $InI_3 > InI$ (b) $AlCl > AlCl_3$

(c) $TlI > TlI_3$ (d) $TlCl_3 > TlCl$

***72.** Some tranquilisers are listed below. Which one from the following belongs to barbiturates?

(a) Meprobamate (b) Valium

(c) Veronal (d) Chlordiazepoxide

73. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Helium is used to dilute oxygen in diving apparatus.

Reason (R) Helium has high solubility in O_2.

In the light of the above statements, choose the correct answer from the options given below:

(a) Both A and R are true and R is not the correct explanation of A.

(b) A is true but R is false.

(c) A is false but R is true.

(d) Both A and R are true and R is the correct explanation of A.

74. The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)

$$[CaCO_3 \xrightarrow{1200\ K} CaO + CO_2]$$

(a) 1.76 g (b) 2.64 g (c) 1.32 g (d) 1.12 g

75. In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe^{3+} due to formation of

(a) NaSCN (b) $[Fe(CN)_5NOS]^{4-}$

(c) $[Fe(SCN)]^{2+}$ (d) $Fe_4[Fe(CN)_6]_3 \cdot xH_2O$

76. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) A reaction can have zero activation energy.

Reason (R) The minimum extra amount of energy absorbed by reactant molecules so that their energy becomes equal to threshold value, is called activation energy.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true and R is not the correct explanation of A.
(b) A is true but R is false.
(c) A is false but R is true.
(d) Both A and R are true and R is the correct explanation of A.

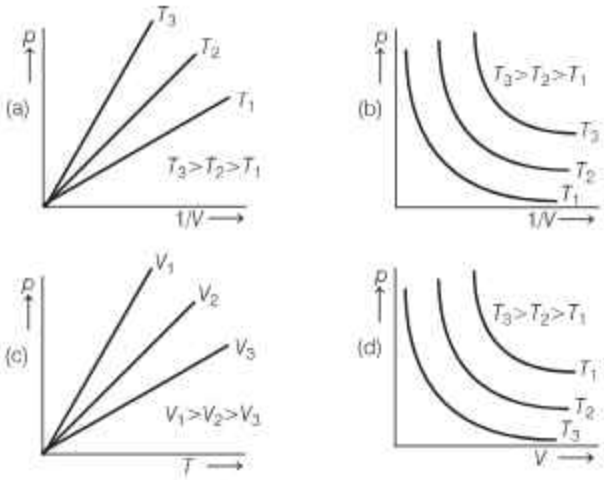
***77.** A compound is formed by two elements *A* and *B*. The element *B* forms cubic close packed structure and atoms of *A* occupy 1/3 of tetrahedral voids. If the formula of the compound is A_xB_y, then the value of $x + y$ is in option
(a) 4 (b) 3
(c) 2 (d) 5

78. The stability of Cu^{2+} is more than Cu^{+} salts in aqueous solution due to
(a) enthalpy of atomisation (b) hydration energy
(c) second ionisation enthalpy
(d) first ionisation enthalpy

79. Select the correct statements from the following:
A. Atoms of all elements are composed of two fundamental particles.
B. The mass of the electrons is 9.10939×10^{-31} kg.
C. All the isotopes of a given element show same chemical properties.
D. Protons and electrons are collectively known as nucleons.
E. Dalton's atomic theory, regarded the atom as an ultimate particle of matter.

Choose the correct answer from the options given below:
(a) C, D and E only (b) A and E only
(c) B, C and E only (d) A, B and C only

***80.** Which amongst the following options is correct graphical representation of Boyle's Law?



81. Intermolecular forces are forces of attraction and repulsion between interacting particles that will include:
A. dipole-dipole forces
B. dipole-induced dipole forces
C. hydrogen bonding
D. covalent bonding
E. dispersion forces

Choose the most appropriate answer from the options given below
(a) A, B, C , D are correct. (b) A, B, C , E are correct.
(c) A, C, D, E are correct. (d) B, C, D, E are correct.

82. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is
(a) 32 (b) 30 (c) 18 (d) 16

***83.** Which one is an example of heterogeneous catalysis?
(a) Hydrolysis of sugar catalysed by H^+ ions.
(b) Decomposition of ozone in presence of nitrogen monoxide.
(c) Combination between dinitrogen and dihydrogen to form ammonia in the presence of finely divided iron.
(d) Oxidation of sulphur dioxide into sulphur trioxide in the presence of oxides of nitrogen.

***84.** Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Metallic sodium dissolves in liquid ammonia giving a deep blue solution, which is paramagnetic.

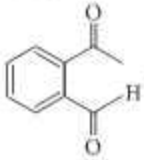
Reason (R) The deep blue solution is due to the formation of amide.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true and R is not the correct explanation of A.
(b) A is true but R is false.
(c) A is false but R is true.
(d) Both A and R are true and R is the correct explanation of A.

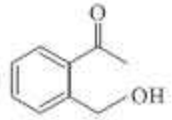
85. Amongst the following, the total number of species not having eight electrons around central atom in its outer most shell, is
NH_3, $AlCl_3$, $BeCl_2$, CCl_4, PCl_5
(a) 2 (b) 4 (c) 1 (d) 3

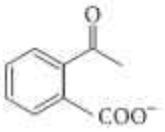
Section-B

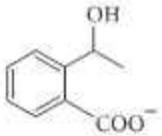
86. Identify the major product obtained in the following reaction

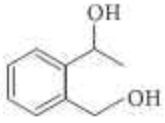


$+ 2[Ag(NH_3)_2]^+ + 3OH^- \xrightarrow{\Delta}$ Major product

(a) 

(b) 

(c) 

(d) 

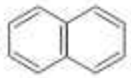
87. Match List-I with List-II.

	List-I (Oxoacids of sulphur)		List-II (Bonds)
A.	Peroxodisulphuric acid	I.	Two S—OH, four S=O, one S—O—S
B.	Sulphuric acid	II.	Two S—OH, one S=O
C.	Pyrosulphuric acid	III.	Two S—OH, Four S=O, one S—O—O—S
D.	Sulphurous acid	IV.	Two S—OH, two S=O

Choose the correct answer from the options given below.

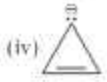
(a) A-III, B-IV, C-I, D-II
(b) A-I, B-III, C-IV, D-II
(c) A-III, B-IV, C-II, D-I
(d) A-I, B-III, C-II, D-IV

88. Consider the following compounds/species.

(i) 

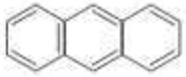
(ii) 

(iii) 

(iv) 

(v) 

(vi) 

(iv) 

The number of compounds/species which obey Huckel's rule is _____ .

(a) 6 (b) 2
(c) 5 (d) 4

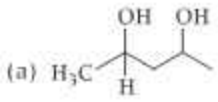
***89.** Pumice stone is an example of

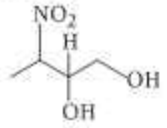
(a) gel (b) solid sol
(c) foam (d) sol

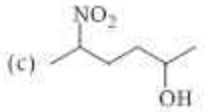
90. Which amongst the following options is the correct relation between change in enthalpy and change in internal energy?

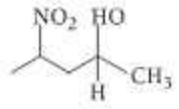
(a) $\Delta H = \Delta U + \Delta n_g RT$
(b) $\Delta H - \Delta U = -\Delta nRT$
(c) $\Delta H + \Delta U = \Delta nR$
(d) $\Delta H = \Delta U - \Delta n_g RT$

91. Which amongst the following will be most readily dehydrated under acidic conditions?

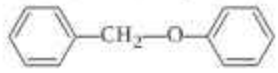
(a) 

(b) 

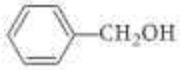
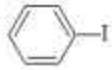
(c) 

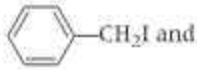
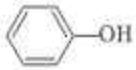
(d) 

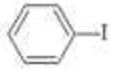
92. Consider the following reaction.

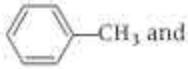
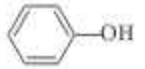


Identify products *A* and *B*.

(a) *A* =  and *B* = 

(b) *A* =  and *B* = 

(c) *A* =  and *B* = 

(d) *A* =  and *B* = 

93. On balancing the given redox reaction,

$$a\mathrm{Cr_2O_7^{2-}} + b\mathrm{SO_3^{2-}}(aq) + c\mathrm{H^+}(aq) \longrightarrow 2a\mathrm{Cr^{3+}}(aq) + b\mathrm{SO_4^{2-}}(aq) + \frac{c}{2}\mathrm{H_2O}(l)$$

the coefficients *a*, *b* and *c* are found to be, respectively

(a) 3, 8, 1 (b) 1, 8, 3 (c) 8, 1, 3 (d) 1, 3, 8

94. Which complex compound is most stable?

(a) $[Co(NH_3)_3(NO_3)_3]$ (b) $[CoCl_2(en)_2]NO_3$
(c) $[Co(NH_3)_6]_2(SO_4)_3$ (d) $[Co(NH_3)_4(H_2O)Br](NO_3)_2$

***95.** What fraction of one edge centred octahedral void lies in one unit cell of *fcc*?

(a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{12}$ (d) $\frac{1}{2}$

96. Which of the following statements are incorrect?

A. All the transition metals except scandium form *MO* oxides which are ionic.
B. The highest oxidation number corresponding to the group number in transition metal oxides is attained in Sc_2O_3 to Mn_2O_7.
C. Basic character increases from V_2O_3 to V_2O_4 to V_2O_5.
D. V_2O_4 dissolves in acids to give VO_4^{3-} salts.
E. CrO is basic but Cr_2O_3 is amphoteric.

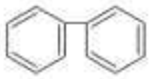
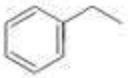
Choose the correct answer from the options given below:

(a) B and D only (b) C and D only
(c) B and C only (d) A and E only

97. The equilibrium concentrations of the species in the reaction, $A + B \rightleftharpoons C + D$ are 2, 3, 10 and 6 mol L^{-1}, respectively at 300 K, ΔG° for the reaction is ($R = 2$ cal/ mol K)
(a) –137.26 cal
(b) –1381.80 cal
(c) –13.73 cal
(d) 1372.60 cal

98. Identify the final product [D] obtained in the following sequence of reactions.

$$CH_3CHO \xrightarrow[(ii)\ H_3O^+]{(i)\ LiAlH_4} [A] \xrightarrow[\Delta]{H_2SO_4} [B] \xrightarrow{HBr} [C] \xrightarrow[Na/dry\ ether]{Br} [D]$$

(a) 
(b) C_4H_{10}
(c) $HC \equiv C^-Na^+$
(d) 

99. Given below are two statements.

Statement I The nutrient deficient water bodies lead to eutrophication.

Statement II Eutrophication leads to decrease in the level of oxygen in the water bodies.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both statement I and statement II are false.
(b) Statement I is true but statement II is false.
(c) Statement I is false but statement II is true.
(d) Both statement I and statement II are true.

***100.** The reaction that does not take place in a blast furnace between 900 K to 1500 K temperature range during extraction of iron is
(a) $FeO + CO \longrightarrow Fe + CO_2$
(b) $C + CO_2 \longrightarrow 2CO$
(c) $CaO + SiO_2 \longrightarrow CaSiO_3$
(d) $Fe_2O_3 + CO \longrightarrow 2FeO + CO_2$

BOTANY

Section-A

101. Among eukaryotes, replication of DNA takes place in
(a) S-phase (b) G_1-phase (c) G_2-phase (d) M-phase

102. Cellulose does not form blue colour with iodine because
(a) it is a helical molecule
(b) it does not contain complex helices and hence cannot hold iodine molecules
(c) it breaks down when iodine reacts with it
(d) it is a disaccharide

103. In gene gun method used to introduce alien DNA into host cells, microparticles of ..... metal are used.
(a) zinc (b) tungsten and gold
(c) silver (d) copper

104. What is the function of tassels in the corn cob?
(a) To trap pollen grains (b) To disperse pollen grains
(c) To protect seeds (d) To attract insects

105. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Late wood has fewer xylary elements with narrow vessels.

Reason (R) Cambium is less active in winters.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true but R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

***106.** The historic convention on biological diversity, 'The Earth Summit' was held in Rio de Janeiro in the year:
(a) 1992 (b) 1986
(c) 2002 (d) 1985

107. What is the role of RNA polymerase-III in the process of transcription in eukaryotes?
(a) Transcription of *t*RNA, 5 *sr*RNA and snRNA
(b) Transcription of precursor of *m*RNA
(c) Transcription of only *sn*RNAs
(d) Transcription of *r*RNAs (28S, 18S and 5,8S)

108. Identify the pair of heterosporous pteridophytes among the following
(a) *Selaginella* and *Salvinia*
(b) *Psilotum* and *Salvinia*
(c) *Equisetum* and *Salvinia*
(d) *Lycopodium* and *Selaginella*

***109.** Given below are two statements.

Statement I The forces generated by transpiration can lift a xylem-sized column of water over 130 metres height.

Statement II Transpiration cools leaf surfaces sometimes 10 to 15 degrees, by evaporative cooling.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both statement I and statement II are incorrect
(b) Statement I is correct, but statement II is incorrect
(c) Statement I is incorrect, but statement II is correct
(d) Both statement I and statement II are correct

110. The reaction centre in PS-II has an absorption maxima at
(a) 700 nm (b) 660 nm (c) 780 nm (d) 680 nm

111. Among 'The Evil Quartet', which one is considered the most important cause driving extinction of species?
(a) Over exploitation for economic gain
(b) Alien species invasions
(c) Co-extinctions
(d) Habitat loss and fragmentation

112. Which of the following stages of meiosis involves division of centromere?
(a) Metaphase-II (b) Anaphase-II
(c) Telophase (d) Metaphase-I

113. Spraying of which of the following phytohormone on juvenile conifers helps in hastening the maturity period, that leads to early seed production?
(a) Gibberellic acid (b) Zeatin
(c) Abscisic acid (d) Indole-3-butyric acid

114. Identify the correct statements.
A. Detritivores perform fragmentation.
B. The humus is further degraded by some microbes during mineralisation.
C. Water soluble inorganic nutrients go down into the soil and get precipitated by a process called leaching.
D. The detritus food chain begins with living organisms.
E. Earthworms breakdown detritus into smaller particles by a process called catabolism.
Choose the correct answer from the options given below.
(a) B, C, D only (b) C, D, E only
(c) D, E, A only (d) A, B, C only

***115.** Which micronutrient is required for splitting of water molecule during photosynthesis?
(a) Molybdenum (b) Magnesium
(c) Copper (d) Manganese

116. Large, colourful, fragrant flowers with nectar are seen in
(a) bird pollinated plants (b) bat pollinated plants
(c) wind pollinated plants (d) insect pollinated plants

***117.** Movement and accumulation of ions across a membrane against their concentration gradient can be explained by
(a) facilitated diffusion (b) passive transport
(c) active transport (d) osmosis

***118.** The thickness of ozone in a column of air in the atmosphere is measured in terms of
(a) decibels (b) decameter
(c) kilobase (d) Dobson units

119. Upon exposure to UV radiation, DNA stained with ethidium bromide will show
(a) bright blue colour (b) bright yellow colour
(c) bright orange colour (d) bright red colour

120. Unequivocal proof that DNA is the genetic material was first proposed by
(a) Alfred Hershey and Martha Chase
(b) Avery, MacLeoid and McCarty
(c) Wilkins and Franklin
(d) Frederick Griffith

121. In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as
(a) dedifferentiation (b) development
(c) senescence (d) differentiation

122. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) ATP is used at two steps in glycolysis.
Reason (R) First ATP is used in converting glucose into glucose-6-phosphate and second ATP is used in conversion of fructose-6-phosphate into fructose-1,6-biphosphate.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true, R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

123. The process of appearance of recombination nodules occurs at which sub-stage of prophase-I in meiosis?
(a) Pachytene (b) Diplotene
(c) Diakinesis (d) Zygotene

124. In the equation GPP $- R =$ NPP
GPP is Gross Primary Productivity.
NPP is Net Primary Productivity. R here is ..... .
(a) Respiratory quotient
(b) Respiratory loss
(c) Reproductive allocation
(d) Photosynthetically active radiation

125. The phenomenon of pleiotropism refers to
(a) presence of two alleles, each of two genes controlling a single trait
(b) a single gene affecting multiple phenotypic expression
(c) more than two genes affecting a single character
(d) presence of several alleles of a single gene controlling a single cross-over

126. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out
(a) DNA (b) histones
(c) polysaccharides (d) RNA

127. In angiosperm, the haploid, diploid and triploid structures of a fertilised embryo sac sequentially are
(a) antipodals, synergids and primary endosperm nucleus
(b) synergids, zygote and primary endosperm nucleus
(c) synergids, antipodanls and polar nuclei
(d) synergids, primary endosperm nucleus and zygote

128. Axile placentation is observed in
(a) China rose, beans and lupin
(b) Tomato, *Dianthus* and Pea
(c) China rose, *Petunia* and lemon
(d) Mustard, cucumber and primrose

129. Which hormone promotes internode/petiole elongation in deep water rice?
(a) Kinetin (b) Ethylene
(c) 2, 4-D (d) GA_3

130. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) The first stage of gametophyte in the life cycle of moss is protonema stage.

Reason (R) Protonema develops directly from spores produced in capsule.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both A and R are true, but R is not the correct explanation of A
(b) A is correct, but R is not correct
(c) A is not correct, but R is correct
(d) Both A and R are correct and R is the correct explanation of A

131. Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by
(a) Sutton and Boveri (b) Alfred Sturtevant
(c) Henking (d) Thomas Hunt Morgan

132. Family-Fabaceae differs from Solanaceae and Liliaceae. With respect to the stamens pick out the characteristics specific to family-Fabaceae, but not found in Solanaceae or Liliaceae.
(a) Polyadelphous and epipetalous stamens
(b) Monoadelphous and monothecous anthers
(c) Epiphyllous and dithecous anthers
(d) Diadelphous and dithecous anthers

133. How many ATP and $NADPH_2$ are required for the synthesis of one molecule of glucose during Calvin cycle?
(a) 18 ATP and 12 $NADPH_2$ (b) 12 ATP and 16 $NADPH_2$
(c) 18 ATP and 16 $NADPH_2$ (d) 12 ATP and 12 $NADPH_2$

134. Expressed Sequence Tags (ESTs) refers to
(a) All genes that are expressed as proteins
(b) All genes whether expressed or unexpressed
(c) Certain important expressed genes
(d) All genes that are expressed as RNA

135. Given below are two statements.

Statement I Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

Statement II Exarch condition is the most common feature of the root system.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both Statement I and Statement II are false
(b) Statement I is correct, but Statement II is false
(c) Statement I is incorrect, but Statement II is true
(d) Both Statement I and Statement II are true

Section-B

136. Which of the following combination is required for chemiosmosis?
(a) Membrane, proton pump, proton gradient, NADP synthase
(b) Proton pump, electron gradient, ATP synthase
(c) Proton pump, electron gradient, NADP synthase
(d) Membrane, proton pump, proton gradient, ATP synthase

137. How many different proteins does the ribosome consist of?
(a) 60 (b) 40 (c) 20 (d) 80

138. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason (R) Internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both A and R are true, but R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

139. Match List-I with List-II.

	List-I		List-II
A.	M-Phase	I.	Proteins are synthesised
B.	G_2-Phase	II.	Inactive phase
C.	Quiescent stage	III.	Interval between mitosis and initiation of DNA replication
D.	G_1-Phase	IV.	Equational division

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-I, D-III (b) A-IV, B-I, C-II, D-III
(c) A-II, B-IV, C-I, D-III (d) A-III, B-II, C-IV, D-I

140. Main steps in the formation of recombinant DNA are given below. Arrange these steps in a correct sequence.
A. Insertion of recombinant DNA into the host cell.
B. Cutting of DNA at specific location by restriction enzyme.
C. Isolation of desired DNA fragment.
D. Amplification of gene of interest using PCR.

Choose the correct answer from the options given below.
(a) C, A, B, D (b) C, B, D, A
(c) B, D, A, C (d) B, C, D, A

141. Given below are two statements.:

Statement I Gause's 'Competitive exclusion principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II In general, carnivores are more adversely affected by competition than herbivores

In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are false
(b) Statement I is true, but Statement II is false
(c) Statement I is false, but Statement II is true
(d) Both Statement I and Statement II are true

***142.** Match List-I with List-II.

	List-I		List-II
A.	Cohesion	I.	More attraction in liquid phase
B.	Adhesion	II.	Mutual attraction among water molecules
C.	Surface tension	III.	Water loss in liquid phase
D.	Guttation	IV.	Attraction towards polar surfaces

Choose the correct answer from the options given below.
(a) A-IV, B-III, C-II, D-I (b) A-III, B-I, C-IV, D-II
(c) A-II, B-I, C-IV, D-III (d) A-II, B-IV, C-I, D-III

143. Melonate inhibits the growth of pathogenic bacteria by inhibiting the activity of
(a) amylase (b) lipase
(c) dinitrogenase (d) succinic dehydrogenase

144. Match List-I with List-II.

	List-I		List-II
A.	Oxidative decarboxylation	I.	Citrate synthase
B.	Glycolysis	II.	Pyruvate dehydrogenase
C.	Oxidative phosphorylation	III.	Electron transport system
D.	Tricarboxylic acid cycle	IV.	EMP pathway

Choose the correct answer from the options given below.
(a) A-II, B-IV, C-I, D-III (b) A-III, B-I, C-II, D-IV
(c) A-II, B-IV, C-III, D-I (d) A-III, B-IV, C-II, D-I

145. Which of the following statements are correct about Klinefelter's syndrome?
A. This disorder was first described by Langdon Down (1866).
B. Such an individual has overall masculine development. However, the feminine development is also expressed.
C. The affected individual is short statured.
D. Physical, psychomotor and mental development is retarded.
E. Such individuals are sterile.

Choose the correct answer from the options given below.
(a) C and D only (b) B and E only
(c) A and E only (d) A and B only

***146.** Which one of the following statements is 'not' correct?
(a) Algal blooms caused by excess of organic matter in water improve water quality and promote fisheries
(b) Water hyacinth grows abundantly in eutrophic water bodies and leads to an imbalance in the ecosystem dynamics of the water body
(c) The amount of some toxic substances of industrial wastewater increases in the organisms at successive trophic levels
(d) The microorganisms involved in biodegradation of organic matter in a sewage polluted water body consume a lot of oxygen causing the death of aquatic organisms

147. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) In gymnosperms, the pollen grains are released from the microsporangium and carried by air currents.

Reason (R) Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true, R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

148. Match List-I with List-II.

	List-I		List-II
A.	Iron	I.	Synthesis of auxin
B.	Zinc	II.	Component of nitrate reductase
C.	Boron	III.	Activator of catalase
D.	Molybdenum	IV.	Cell elongation and differentiation

Choose the correct answer from the options given below.
(a) A-II, B-III, C-IV, D-I (b) A-III, B-I, C-IV, D-II
(c) A-II, B-IV, C-I, D-III (d) A-III, B-II, C-I, D-IV

149. Match List-I with List-II.

	List-I (Interaction)	List-II (Species *A* and *B*)
A.	Mutualism	I. $+(A), 0(B)$
B.	Commensalism	II. $-(A), 0(B)$
C.	Amensalism	III. $+(A), -(B)$
D.	Parasitism	IV. $+(A), +(B)$

Choose the correct answer from the options given below.
(a) A-IV, B-I, C-II, D-III (b) A-IV, B-III, C-I, D-II
(c) A-III, B-I, C-IV, D-II (d) A-IV, B-II, C-I, D-III

150. Identify the correct statements.
A. Lenticels are the lens-shaped openings permitting the exchange of gases.
B. Bark formed early in the season is called hard bark.
C. Bark is a technical term that refers to all tissues exterior to vascular cambium.
D. Bark refers to periderm and secondary phloem.
E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below.
(a) A and D only (b) A, B and D only
(c) B and C only (d) B, C and E only

ZOOLOGY

Section-A

***151.** Once the undigested and unabsorbed substances enter the caecum, their backflow is prevented by
(a) ileo-caecal valve
(b) gastro - oesophageal sphincter
(c) pyloric sphincter
(d) sphincter of Oddi

152. In which blood corpuscles, the HIV undergoes replication and produces progeny viruses?
(a) B-lymphocytes (b) Basophils
(c) Eosinophils (d) T_H-cells

153. Broad palm with single palm crease is visible in a person suffering from
(a) Turner's syndrome (b) Klinefelter's syndrome
(c) thalassemia (d) Down's syndrome

154. Which one of the following common sexually transmitted diseases is completely curable when detected early and treated properly?
(a) Gonorrhoea (b) Hepatitis-B
(c) HIV infection (d) Genital herpes

155. Match List-I with List-II.

	List-I (Interacting species)		**List-II** (Name of interaction)
A.	A leopard and a lion in a forest/grassland	I.	Competition
B.	A cuckoo laying egg in a crow's nest	II.	Brood parasitism
C.	Fungi and root of a higher plant in mycorrhizae	III.	Mutualism
D.	A cattle egret and a Cattle in a field	IV.	Commensalism

Choose the correct answer from the options given below.
(a) A-I, B-II, C-IV, D-III
(b) A-III, B-IV, C-I, D-II
(c) A-II, B-III, C-I, D-IV
(d) A-I, B-II, C-III, D-IV

***156.** Match List-I with List-II.

	List-I (Cells)		**List-II** (Secretion)
A.	Peptic cells	I.	Mucus
B.	Goblet cells	II.	Bile juice
C.	Oxyntic cells	III.	Proenzyme pepsinogen
D.	Hepatic cells	IV.	HCl and intrinsic factor for absorption of vitamin-B_{12}

Choose the correct answer from the options given below.
(a) A-II, B-I, C-III, D-IV (b) A-III, B-I, C-IV, D-II
(c) A-II, B-IV, C-I, D-III (d) A-IV, B-III, C-II, D-I

157. Given below are two statements:

Statement I A protein is imagined as a line, the left end represented by first amino acid (C-terminal) and the right end represented by last amino acid (N-terminal).

Statement II Adult human haemoglobin, consists of 4 subunits (two subunits of α type and two subunits of β type).

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are false
(b) Statement I is true, but statement II is false
(c) Statement I is false, but statement II is true
(d) Both statement I and statement II are true

158. Given below are two statements:

Statement I RNA mutates at a faster rate.

Statement II Viruses having RNA genome and shorter lifespan mutate and evolve faster.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are false
(b) Statement I is true, but Statement II is false
(c) Statement I is false, but Statement II is true
(d) Both Statement I and Statement II are true

159. Match List-I with List-II.

	List-I		List-II
A.	Vasectomy	I.	Oral method
B.	Coitus interruptus	II.	Barrier method
C.	Cervical caps	III.	Surgical method
D.	Saheli	IV.	Natural method

Choose the correct answer from the options given below.
(a) A-III, B-IV, C-II, D-I (b) A-II, B-III, C-I, D-IV
(c) A-IV, B-II, C-I, D-III (d) A-III, B-I, C-IV, D-II

160. Match List-I with List-II.

	List-I		List-II
A.	Ringworm	I.	*Haemophilus influenzae*
B.	Filariasis	II.	*Trichophyton*
C.	Malaria	III.	*Wuchereria bancrofti*
D.	Pneumonia	IV.	*Plasmodium vivax*

Choose the correct answer from the options given below.
(a) A-II, B-III, C-I, D-IV (b) A-III, B-II, C-I, D-IV
(c) A-III, B-II, C-IV, D-I (d) A-II, B-III, C-IV, D-I

161. Match List-I with List-II.

	List-I		List-II
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the correct answer from the options given below.
(a) A-III, B-II, C-IV, D-I (b) A-II, B-IV, C-I, D-III
(c) A-IV, B-II, C-III, D-I (d) A-IV, B-III, C-II, D-I

162. Given below are two statements.

Statement I Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

Statement II The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are false
(b) Statement I is true, but statement II is false
(c) Statement I is false, but statement II is true
(d) Both statement I and statement II are true

163. Which one of the following techniques does not serve the purpose of early diagnosis of a disease for its early treatment?
(a) Serum and urine analysis
(b) Polymerase Chain Reaction (PCR) technique
(c) Enzyme Linked Immuno-Sorbent Assay (ELISA) technique
(d) Recombinant DNA technology

164. Which of the following are not considered as the part of endomembrane system?
A. Mitochondria B. Endoplasmic reticulum
C. Chloroplast D. Golgi complex
E. Peroxisomes
Choose the most appropriate answer from the options given below.
(a) A, C and E only (b) A and D only
(c) A, D and E only (d) B and D only

165. Match List-I with List-II.

	List-I		List-II
A.	Heroin	I.	Effect on cardiovascular system
B.	Marijuana	II.	Slow down body function
C.	Cocaine	III.	Painkiller
D.	Morphine	IV.	Interfere with transport of dopamine

Choose the correct answer from the options given below.
(a) A-I, B-II, C-III, D-IV (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-I, D-II (d) A-II, B-I, C-IV, D-III

166. Given below are two statements.

Statement I Low temperature preserves the enzyme in a temporarily inactive state whereas high temperature destroys enzymatic activity because proteins are denatured by heat.

Statement II When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are false
(b) Statement I is true, but statement II is false
(c) Statement I is false, but statement II is true
(d) Both statement I and statement II are true

167. Which of the following functions is carried out by cytoskeleton in a cell?
(a) Protein synthesis (b) Motility
(c) Transportation (d) Nuclear division

168. Given below are two statements.

Statement I Ligaments are dense irregular tissue.

Statement II Cartilage is dense regular tissue.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and statement II are false
(b) Statement I is true, but statement II is false
(c) Statement I is false, but statement II is true
(d) Both statement I and statement II are true

169. Match List-I with List-II.

	List-I		List-II
A.	Cartilaginous joint	I.	Between flat skull bones
B.	Ball and socket joint	II.	Between adjacent vertebrae in vertebral column
C.	Fibrous joint	III.	Between carpal and metacarpal of thumb
D.	Saddle joint	IV.	Between humerus and pectoral girdle

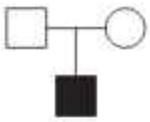
Choose the correct answer from the options given below.
(a) A-II, B-IV, C-I, D-III
(b) A-I, B-IV, C-III, D-II
(c) A-II, B-IV, C-III, D-I
(d) A-III, B-I, C-II, D-IV

170. Match List-I with List-II.

	List-I		List-II
A.	Gene *'a'*	I.	β-galactosidase
B.	Gene *'y'*	II.	Transacetylase
C.	Gene *'i'*	III.	Permease
D.	Gene *'z'*	IV.	Repressor protein

Choose the correct answer from the options given below.
(a) A-II, B-III, C-IV, D-I (b) A-III, B-IV, C-I, D-II
(c) A-III, B-I, C-IV, D-II (d) A-II, B-I, C-IV, D-III

171. Which one of the following symbols represents mating between relative in human pedigree analysis?

(a) 
(b) 
(c) 
(d) 

172. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Endometrium is necessary for implantation of blastocyst.

Reason (R) In the absence of fertilisation, the corpus luteum degenerates that causes disintegration of endometrium.

In the light of the above statements, choose thecorrect answer from the options given below.
(a) Both A and R are true, but R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

173. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Nephrons are of two types: Cortical and juxta medullary, based on their relative position in cortex and medulla.

Reason (R) Juxta medullary nephrons have short loop of Henle whereas, cortical nephrons have longer loop of Henle.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true, but R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

174. Radial symmetry is not found in adults of phylum
(a) Hemichordata (b) Coelenterata
(c) Echinodermata (d) Ctenophora

***175.** Match List-I with List-II.

	List-I		List-II
A.	Fovea	I.	Visible coloured portion of eye that regulates diameter of pupil.
B.	Iris	II.	External layer of eye formed of dense connective tissue.
C.	Blind spot	III.	Point of greatest visual acuity or resolution.
D.	Sclera	IV.	Point where optic nerve leaves the eyeball and photoreceptor cells are absent.

Choose the correct answer from the options given below.
(a) A-IV, B-III, C-II, D-I (b) A-I, B-IV, C-III, D-II
(c) A-II, B-I, C-III, D-IV (d) A-III, B-I, C-IV, D-II

***176.** Which of the following statements is correct?
(a) Biomagnification refers to increase in concentration of the toxicant at successive trophic levels
(b) Presence of large amount of nutrients in water restricts 'algal bloom'
(c) Algal bloom decreases fish mortality
(d) Eutrophication refers to increase in domestic sewage and wastewater in lakes

177. Which of the following statements are correct regarding female reproductive cycle?
A. In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
B. First menstrual cycle begins at puberty and is called menopause.
C. Lack of menstruation may be indicative of pregnancy.
D. Cyclic menstruation extends between menarche and menopause.

Choose the most appropriate answer from the options given below.
(a) A and B only (b) A, B and C only
(c) A, C and D only (d) A and D only

178. Given below are two statements.

Statement I In prokaryotes, the positively charged DNA is held with some negatively charged proteins in a region called nucleoid.

Statement II In eukaryotes, the negatively charged DNA is wrapped around the positively charged histone octamer to form nucleosome.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both statement I and statement II are false
(b) Statement I is true, but statement II is false
(c) Statement I is false, but statement II is true
(d) Both statement I and statement II are true

179. Select the correct group/set of Australian Marsupials exhibiting adaptive radiation.
(a) Numbat, spotted cuscus, flying phalanger
(b) Mole, flying squirrel, Tasmanian tiger cat
(c) Lemur, anteater, wolf
(d) Tasmanian wolf, bobcat, marsupial mole

180. Match List-I with List-II.

	List-I		List-II
A.	*Taenia*	I.	Nephridia
B.	*Paramecium*	II.	Contractile vacuole
C.	*Periplaneta*	III.	Flame cells
D.	*Pheretima*	IV.	Urecose gland

Choose the correct answer from the options given below.
(a) A-I, B-II, C-IV, D-III (b) A-III, B-II, C-IV, D-I
(c) A-II, B-I, C-IV, D-III (d) A-I, B-II, C-III, D-IV

181. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) Amniocentesis for sex determination is one of the strategies of reproductive and child healthcare programme.

Reason (R) Ban on amniocentesis checks increasing menace of female foeticide.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true, but R is not the correct explanation of A
(b) A is true, but R is false
(c) A is false, but R is true
(d) Both A and R are true and R is the correct explanation of A

182. Which of the following is not a cloning vector?
(a) YAC (b) pBR322
(c) Probe (d) BAC

***183.** Given below are two statements.

Statement I Electrostatic precipitator is most widely used in thermal power plant.

Statement II Electrostatic precipitator in thermal power plant removes ionising radiations.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are false
(b) Statement I is true, but Statement II is incorrect
(c) Statement I is false, but Statement II is correct
(d) Both Statement I and Statement II are true

184. Match List-I with List-II.

	List-I		List-II
A.	P-wave	I.	Beginning of systole
B.	Q-wave	II.	Repolarisation of ventricles
C.	QRS complex	III.	Depolarisation of atria
D.	T-wave	IV.	Depolarisation of ventricles

Choose the correct answer from the options given below.
(a) A-IV, B-III, C-II, D-I (b) A-II, B-IV, C-I, D-III
(c) A-I, B-II, C-III, D-IV (d) A-III, B-I, C-IV, D-II

185. Vital capacity of lung is
(a) IRV + ERV + TV + RV (b) IRV + ERV + TV – RV
(c) IRV + ERV +TV (d) IRV + ERV

Section-B

186. Given below are two statements.

Statement I During G_0-phase of cell cycle, the cell is metabolically inactive.

Statement II The centrosome undergoes duplication during S-phase of interphase.

In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both Statement I and Statement II are false
(b) Statement I is true, but Statement II is false
(c) Statement I is false, but Statement II is true
(d) Both Statement I and Statement II are true

187. The parts of human brain that helps in regulation of sexual behaviour, expression of excitement, pleasure, rage, fear, etc. are
(a) corpora quadrigemina and hippocampus
(b) brain stem and epithalamus
(c) corpus callosum and thalamus
(d) limbic system and hypothalamus

188. In cockroach, excretion is brought about by –
A. Phallic gland B. Urecose gland
C. Nephrocytes D. Fat body
E. Collaterial glands
Choose the correct answer from the options given below.
(a) A, B and E only (b) B, C and D only
(c) B and D only (d) A and E only

189. Which of the following statements are correct?

A. An excessive loss of body fluid from the body switches off osmoreceptors.

B. ADH facilitates water reabsorption to prevent diuresis.

C. ANF causes vasodilation.

D. ADH causes increase in blood pressure.

E. ADH is responsible for decrease in GFR.

Choose the correct answer from the options given below.

(a) B, C and D only (b) A, B and E only

(c) C, D and E only (d) A and B only

***190.** Which one of the following is not an advantage of inbreeding?

(a) It exposes harmful recessive genes that are eliminated by selection

(b) Elimination of less desirable genes and accumulation of superior genes takes place due to it

(c) It decreases the productivity of inbreed population, after continuous inbreeding

(d) It decreases homozygosity

191. Which of the following statements are correct regarding skeletal muscle?

A. Muscle bundles are held together by collagenous connective tissue layer called fascicle.

B. Sarcoplasmic reticulum of muscle fibre is a store house of calcium ions.

C. Striated appearance of skeletal muscle fibre is due to distribution pattern of actin and myosin proteins.

D. M-line is considered as functional unit of contraction called sarcomere.

Choose the most appropriate answer from the options given below.

(a) B and C only (b) A, C and D only

(c) C and D only (d) A, B and C only

192. Match List-I with List-II.

	List-I		List-II
A.	Logistic growth	I.	Unlimited resource availability condition
B.	Exponential growth	II.	Limited resource availability condition
C.	Expanding age pyramid	III.	The per cent individuals of pre-reproductive age is largest followed by reproductive and post-reproductive age groups
D.	Stable age pyramid	IV.	The per cent individuals of pre-reproductive and reproductive age group are same

Choose the correct answer from the options given below.

(a) A-II, B-III, C-I, D-IV (b) A-II, B-IV, C-I, D-III

(c) A-II, B-IV, C-III, D-I (d) A-II, B-I, C-III, D-IV

193. Which one of the following is the sequence on corresponding coding strand, if the sequence on *m*RNA formed is as follows

5'AUCGAUCGAUCGAUCGAUCGAUCG AUCG 3'?

(a) 3' UAGCUAGCUAGCUAGCUAGCUAGCUAGC 5'

(b) 5' ATCGATCGATCGATCGATCGATCGATCG 3'

(c) 3' ATCGATCGATCGATCGATCGATCGATCG 5'

(d) 5' UAGCUAGCUAGCUAGCUAGCUAGCUAGC 3'

194. Which of the following statements are correct.

A. Basophils are most abundant cells of the total WBCs.

B. Basophils secrete histamine, serotonin and heparin.

C. Basophils are involved in inflammatory response.

D. Basophils have kidney shaped nucleus.

E. Basophils are agranulocytes.

Choose the correct answer from the options given below.

(a) C and E only (b) B and C only

(c) A and B only (d) D and E only

195. Match List-I with List-II.

	List-I		List-II
A.	Mast cells	I.	Ciliated epithelium
B.	Inner surface of bronchiole	II.	Areolar connective tissue
C.	Blood	III.	Cuboidal epithelium
D.	Tubular parts of nephron	IV.	Specialised connective tissue

Choose the correct answer from the options given below.

(a) A-II, B-III, C-I, D-IV

(b) A-II, B-I, C-IV, D-III

(c) A-III, B-IV, C-II, D-I

(d) A-I, B-II, C-IV, D-III

196. Which of the following is characteristic feature of cockroach regarding sexual dimorphism?

(a) Presence of anal styles

(b) Presence of sclerites

(c) Presence of anal cerci

(d) Dark brown body colour and anal cerci

197. Which one of the following are not under the control of thyroid hormone?

A. Maintenance of water and electrolyte balance.

B. Regulation of basal metabolic rate.

C. Normal rhythm of sleep-wake cycle.

D. Development of immune system.

E. Support the process of RBCs formation.

Choose the correct answer from the options given below.

(a) B and C only (b) C and D only

(c) D and E only (d) A and D only

198. Select the correct statements.

A. Tetrad formation is seen during leptotene.

B. During anaphase, the centromeres split and chromatids separate.

C. Terminalisation takes place during pachytene.

D. Nucleolus, Golgi complex reformed during telophase.

E. Crossing over takes place between sister chromatids of homologous chromosome.

Choose the correct answer from the options given below.

(a) B and D only (b) A, C and E only
(c) B and E only (d) A and C only

199. The unique mammalian characteristics are

(a) hairs, pinna and mammary glands
(b) hairs, pinna and indirect development
(c) pinna, monocondylic skull and mammary glands
(d) hairs, tympanic membrane and mammary glands

200. Select the correct statements with reference to chordates.

A. Presence of a mid-dorsal, solid and double nerve cord.

B. Presence of closed circulatory system.

C. Presence of paired pharyngeal gill slits.

D. Presence of dorsal heart.

E. Triploblastic pseudocoelomate animals.

Choose the correct answer from the options given below.

(a) B and C only
(b) B, D and E only
(c) C, D and E only
(d) A, C and D only

Answers

Physics

1. (b)	2. (b)	3. (d)	4. (c)	5. (c)	6. (a)	7. (c)	8. (c)	9. (c)	10. (b)
11. (a)	12. (a)	13. (d)	14. (d)	15. (c)	16. (d)	17. (b)	18. (a)	19. (b)	20. (*)
21. (d)	22. (d)	23. (c)	24. (b)	25. (b)	26. (b)	27. (b)	28. (d)	29. (a)	30. (c)
31. (b)	32. (*)	33. (b)	34. (b)	35. (a)	36. (c)	37. (a)	38. (*)	39. (d)	40. (a)
41. (b)	42. (c)	43. (b)	44. (*)	45. (b)	46. (b)	47. (b)	48. (c)	49. (a)	50. (b)

Chemistry

51. (d)	52. (b)	53. (b)	54. (d)	55. (d)	56. (d)	57. (a)	58. (d)	59. (b)	60. (d)
61. (b)	62. (a)	63. (d)	64. (b)	65. (b)	66. (b)	67. (b)	68. (b)	69. (b)	70. (c)
71. (c)	72. (c)	73. (b)	74. (a)	75. (c)	76. (c)	77. (d)	78. (b)	79. (c)	80. (a)
81. (b)	82. (a)	83. (c)	84. (b)	85. (d)	86. (b)	87. (a)	88. (d)	89. (b)	90. (a)
91. (a)	92. (b)	93. (d)	94. (b)	95. (b)	96. (b)	97. (b)	98. (d)	99. (c)	100. (d)

Botany

101. (a)	102. (b)	103. (b)	104. (a)	105. (d)	106. (a)	107. (a)	108. (a)	109. (d)	110. (d)
111. (d)	112. (b)	113. (a)	114. (d)	115. (d)	116. (d)	117. (c)	118. (d)	119. (c)	120. (a)
121. (a)	122. (d)	123. (a)	124. (b)	125. (b)	126. (a)	127. (b)	128. (c)	129. (b)	130. (d)
131. (b)	132. (d)	133. (a)	134. (d)	135. (c)	136. (d)	137. (d)	138. (d)	139. (b)	140. (d)
141. (b)	142. (d)	143. (d)	144. (c)	145. (b)	146. (a)	147. (b)	148. (b)	149. (a)	150. (a)

Zoology

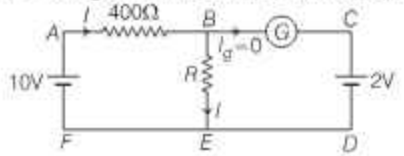
151. (a)	152. (d)	153. (d)	154. (a)	155. (d)	156. (b)	157. (c)	158. (d)	159. (a)	160. (d)
161. (d)	162. (d)	163. (a)	164. (a)	165. (d)	166. (d)	167. (b)	168. (a)	169. (a)	170. (a)
171. (a)	172. (a)	173. (b)	174. (a)	175. (d)	176. (a)	177. (c)	178. (c)	179. (a)	180. (b)
181. (c)	182. (c)	183. (b)	184. (d)	185. (c)	186. (c)	187. (d)	188. (b)	189. (a)	190. (c)
191. (a)	192. (d)	193. (b)	194. (b)	195. (b)	196. (a)	197. (b)	198. (a)	199. (a)	200. (a)

Note *(*) None of the option is correct.*

Answer with Explanations

Physics

1. *(b)* The given situation is shown below.



Applying Kirchhoff's voltage law in the loop *ABCDEFA*, we get

$$400I + 2 - 10 = 0$$

$$\Rightarrow \quad I = \frac{8}{400} = \frac{1}{50}\text{ A}$$

Applying Kirchhoff's voltage law in loop *ABEFA*, we get

$$400I + RI - 10 = 0$$

$$\Rightarrow \quad (400 + R)I = 10$$

$$\Rightarrow \quad (400 + R) \times \frac{1}{50} = 100$$

$$\Rightarrow \quad 400 + R = 500$$

$$\Rightarrow \quad R = 100\,\Omega$$

2. *(b)* When AC is converted into DC using a rectifier circuit, then converted DC is an unstable current that includes ripples. A capacitor is used to remove these ripples and maintain a constant voltage across load.

3. *(d)* Efficiency of carnot engine,

$\eta = 50\% = 0.5$

Temperature of source,

$T_1 = 327°\text{C} = (327 + 273)\text{ K} = 600\text{ K}$

If T_2 be the temperature of sink, then

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \quad 0.5 = 1 - \frac{T_2}{600} \Rightarrow \frac{T_2}{600} = 0.5$$

$$\Rightarrow \quad T_2 = 300\text{ K} = (300 - 273)°\text{C} = 27°\text{C}$$

4. *(c)* According to given situation,

$$c = \frac{x}{t_1} \text{ and } v = \frac{10x}{t_2}$$

Since, $\sin i_c = \frac{1}{\mu} = \frac{1}{(c/v)} = \frac{v}{c}$

$$= \frac{\frac{10x}{t_2}}{\frac{x}{t_1}} = \frac{10t_1}{t_2}$$

$$\Rightarrow \quad i_c = \sin^{-1}\left(\frac{10t_1}{t_2}\right)$$

5. *(c)* The direction of angular acceleration of a body moving along the circumference of the circular path is perpendicular to the plane in which rotation takes place i.e. along the axis of rotation.

6. *(a)* Venturimeter is a device used to measure the flow speed (flow rate) of incompressible fluid. It works on the Bernoulli's principle.

7. *(c)* Given, $T_{1/2} = 20$ minutes

According to given condition,

$$R = \frac{R_0}{16}$$

But, $\quad R = R_0\left(\frac{1}{2}\right)^n$

$$\Rightarrow \quad \frac{R_0}{16} = R_0\left(\frac{1}{2}\right)^n \Rightarrow \left(\frac{1}{2}\right)^4 = \left(\frac{1}{2}\right)^n$$

$$\Rightarrow \quad n = 4 \quad \Rightarrow \frac{t}{T/2} = 4$$

$$\Rightarrow \quad t = 4T_{1/2}$$

$$= 4 \times 20 = 80\text{ minutes}$$

8. *(c)* Random errors can arise due to random and unpredictable fluctuations in temperature and voltage supply.

9. *(c)* Given, $\theta = 30°$

$$E = 2 \times 10^5\text{ NC}^{-1}$$

Dipole length, $2l = 2\text{ cm} = 2 \times 10^{-2}$ m

Torque, $\tau = 4$ N-m

We know that,

$$\Rightarrow \quad \tau = pE \sin\theta$$

$$\Rightarrow \quad p = \frac{\tau}{E \sin\theta} = \frac{4}{2 \times 10^5 \times \sin 30°}$$

$$\Rightarrow \quad p = 4 \times 10^{-5}\text{ C-m}$$

$$\Rightarrow \quad 2l \cdot q = 4 \times 10^{-5}$$

$$\Rightarrow \quad q = \frac{4 \times 10^{-5}}{2l} = \frac{4 \times 10^{-5}}{2 \times 10^{-2}} = 2 \times 10^{-3}\text{ C}$$

$$= 2\text{ mC}$$

10. *(b)* Friction force always works against the motion of the football player. Here, football player is moving in the southward, so friction will act in northword. From the Newton's law of motion, force is in the direction of change in momentum. Since, the football player is just trying to move in the east direction in which no initial momentum was present, so force will be in north-east direction.

11. *(a)* In hydrogen spectrum,

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For shortest wavelength in the Balmer series,

$$n_1 = 2 \text{ and } n_2 = \infty$$

$$\therefore \quad \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$$

$$\Rightarrow \quad \frac{1}{\lambda} = \frac{R}{4}$$

$$\Rightarrow \quad R = \frac{4}{\lambda} \qquad \ldots(\text{i})$$

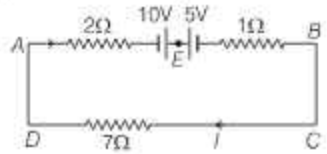
For the shortest wavelength of Bracket series, $n_1 = 4$ and $n_2 = \infty$

$$\therefore \quad \frac{1}{\lambda_B} = R\left(\frac{1}{4^2} - \frac{1}{\infty^2}\right)$$

$$= \frac{R}{16} = \frac{4/\lambda}{16}$$

$$\Rightarrow \quad \frac{1}{\lambda_B} = \frac{1}{4\lambda} \Rightarrow \lambda_B = 4\lambda$$

12. *(a)* The circuit diagram is shown as



Applying KVL in the closed loop *AEBCDA*,

$$2I - 10 + 5 + I \cdot 1 + 7I = 0$$

$$\Rightarrow \quad 10I = 5$$

$$\Rightarrow \quad I = 0.5\text{A}$$

Since, current is positive, hence direction of current is from *A* to *B* through *E*.

13. *(d)* According to given situation,

$$V_S = 12\text{ V},$$

$$\therefore \quad I_S = \frac{P}{V_S} = \frac{60}{12} = 5\text{ A}$$

$\therefore$ Given, $V_P = 220$ V

For ideal transformer,

$$I_P V_P = I_S V_S$$

$$\Rightarrow \quad I_P = \frac{I_S V_S}{V_P} = \frac{5 \times 12}{220} = 0.27\text{ A}$$

14. *(d)* For photoelectric emission, energy of incident electromagnetic radiation must be greater than or equal to work function of photosensitive surface. Thus, only Cs can emit photoelectrons.

15. *(c)* Given, when $x_1 = 2$ cm

Then, $U_1 = U \Rightarrow \frac{1}{2} kx_1^2 = U$

$$\Rightarrow \quad K = \frac{2U}{x_1^2} \qquad \ldots(i)$$

When $x_2 = 8$ cm, then $U = \frac{1}{2} kx_2^2$

$$= \frac{1}{2} \times \frac{2U}{x_1^2} \times x_2^2 \quad \text{[From Eq. (i)]}$$

$$= U \cdot \left(\frac{x_2}{x_1}\right)^2 = U\left(\frac{8}{2}\right)^2 = 16U$$

16. *(d)* Net magnetic flux through a closed surface is zero because the number of field lines leaving the surface are equal to the number of field lines entering the surface.

17. *(b)* Given, surface tension, $T = 0.03\ \text{Nm}^{-1}$

Radius of soop bubble, $r = 2$ cm $= 2 \times 10^{-2}$m

$\therefore$ Amount of energy required,

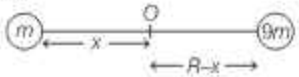
$$E = 2TA = 2 \times 0.03 \times 4\pi (2 \times 10^{-2})^2$$

$$= 3.01 \times 10^{-4}\ \text{J}$$

18. *(a)* Longitudinal stress $= \frac{\text{Weight}}{\text{Area}} = \frac{w}{A}$



19. *(b)* The given situation is shown below.



Since, gravitational field at point O is zero, hence

$$\frac{Gm}{x^2} = \frac{G(9m)}{(R-x)^2}$$

$$\Rightarrow \quad \left(\frac{1}{x}\right)^2 = \left(\frac{3}{R-x}\right)^2$$

$$\Rightarrow \quad \frac{1}{x} = \frac{3}{R-x} \Rightarrow 3x = R - x$$

$$\Rightarrow \quad x = \frac{R}{4}$$

$\therefore$ Gravitational potential at point O,

$$V = \frac{-Gm}{x} + \left[\frac{-G(9m)}{R-x}\right]$$

$$= \frac{-Gm}{\frac{R}{4}} - \frac{9Gm}{R - \frac{R}{4}}$$

$$= -\frac{4Gm}{R} - \frac{12Gm}{R} = -\frac{16Gm}{R}$$

20. (*) Initial temperature of the gas,

$T_1 = -50°\text{C} = (-50 + 273)\text{K} = 223\text{K}$

Final temperature of the gas,

$T_2 = ?$

$(v_{rms})_1 = v$

$(v_{rms})_2 = 3v$

We know that,

$$v_{rms} = \sqrt{\frac{3RT}{m}} \Rightarrow V_{rms} \propto \sqrt{T}$$

$$\Rightarrow \quad \frac{(v_{rms})_2}{(v_{rms})_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow \frac{3v}{v} = \sqrt{\frac{T_2}{223}}$$

$$\Rightarrow \quad 9 = \frac{T_2}{223} \Rightarrow T_2 = 2007\text{K}$$

21. *(d)* Photovoltaic devices can convert optical radiation into electricity. The basic example of such device is solar photovoltaic system which converts solar energy directly into electrical energy.

Zener diode is heavily doped p-n junction diode operating in reverse biased condition in breakdown region.

22. *(d)* According to Gauss's law, $\oint \mathbf{E} \cdot d\mathbf{S}$ represents total electric flux passing through a closed surface.

If $\oint \mathbf{E} \cdot d\mathbf{S} = 0$, then it signifies that the number of electric flux lines entering the surface must be equal to the number of flux lines leaving the surface.

23. *(c)* In series L-C-R circuit,

$L = 10\ \text{mH} = 10^{-2}\ \text{H}$

$C = 1\ \text{F} = 10^{-6}\ \text{F}$

$R = 100\ \Omega$

$\therefore$ Resonance frequency, $f = \frac{1}{2\pi\sqrt{LC}}$

$$= \frac{1}{2\pi\sqrt{10^{-2} \times 10^{-6}}}$$

$$= \frac{1}{2\pi \times 10^{-4}} = \frac{10^4}{2\pi}$$

$$= 1.59 \times 10^3\ \text{Hz} = 1.59\ \text{kHz}$$

24. *(b)* We know that, capacitive reactance,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

i.e., due to decrease in operating frequency (f), X_C increases. As X_C increases, the conduction current decreases.

Since, in case of capacitor, conduction current is equal to displacement current.

Thus, with the decrease of operating frequency, displacement current decreases.

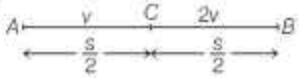
25. *(b)* Given, $u = 280\ \text{ms}^{-1}$, $\theta = 30°$

$\therefore$ Maximum height attaiend by the bullet.

$$H_{max} = \frac{u^2 \sin^2\theta}{2g} = \frac{(280)^2 \sin^2 30°}{2 \times 9.8}$$

$$= \frac{(280)^2 \times (0.5)^2}{2 \times 9.8} = 1000\ \text{m}$$

26. *(b)*



Time taken by vehicle to cover first half,

$$t_1 = \frac{s/2}{v} = \frac{s}{2v}$$

Time taken by vehicle to cover second half,

$$t_2 = \frac{s/2}{2v} = \frac{s}{4v}$$

$\therefore$ Average speed,

$$v_{avg} = \frac{\text{Total distance covered}}{\text{Total time taken}}$$

$$= \frac{\frac{s}{2} + \frac{s}{2}}{t_1 + t_2} = \frac{s}{\frac{s}{2v} + \frac{s}{4v}} = \frac{4v}{3}$$

27. *(b)* Given, $R = (22000 \pm 5\%)\ \Omega$

$$= (22 \times 10^3 \pm 5\%)\ \Omega$$

Here, colours of four bands are

Red	Red	Orange	Gold
↓	↓	↓	↓
2	2	3	± 5%

Thus, colour of third band is orange.

28. *(d)* Equivalent capacitance of the system,

$$C_{AB} = \frac{3 \times (3+3)}{3 + (3+3)} = \frac{18}{9} = 2\ \mu\text{F}$$

29. *(a)* The minimum wavelength of X-rays produced by an electron accelerated through a potential difference V is given as

$$\lambda_{min} = \frac{hc}{eV}$$

$$\Rightarrow \quad \lambda_{min} \propto \frac{1}{V}$$

30. *(c)* Stored magnetic energy,

$$U = \frac{1}{2} LI^2$$

Here, $L = 4\mu H = 4 \times 10^{-6}$ H

$I = 2$ A

$$\therefore \quad U = \frac{1}{2} \times 4 \times 10^{-6} \times 2^2$$

$$= 8 \times 10^{-6} \text{ J} = 8 \mu J$$

31. *(b)* Angular separation of fringes,

$$\theta = \frac{\lambda}{d}$$

i.e., Angular separation does not depend on the distance between screen and plane of slits. Thus, statement I is correct.

Since, $\theta \propto \lambda$

Thus, if the monochromatic source is replaced by another monochromatic source of larger wavelength, then angular seperation of fringes increases. Hence, statement II is incorrect.

32. (*) Radius of gyration of solid sphere and hollow sphere are K_1 and K_2 respectively, then

$$MK_1^2 = \frac{2}{5} MR^2$$

$$\Rightarrow \quad K_1 = \sqrt{\frac{2}{5}} R$$

and $$MK_2^2 = \frac{2}{3} MR^2$$

$$\Rightarrow \quad K_2 = \sqrt{\frac{2}{3}} R$$

$$\therefore \quad \frac{K_1}{K_2} = \frac{\sqrt{2/5}R}{\sqrt{2/3}R} = \sqrt{\frac{3}{5}}$$

33. *(b)* Given, $E_0 = 48$ Vm^{-1}

$$c = 3 \times 10^8 \text{ m/s}$$

$\therefore$ Amplitude of oscillating magnetic field,

$$B_0 = \frac{E_0}{c}$$

$$= \frac{48}{3 \times 10^8} = 1.6 \times 10^{-7} \text{ T}$$

34. *(b)* Given, $M = (0.4 \pm 0.002)$ g

$R = (0.3 \pm 0.001)$ mm

$L = (5 \pm 0.02)$ cm

$\therefore$ Density of wire, $\rho = \frac{M}{\pi r^2 L}$

$\therefore$ Maximum percentage error in the measurement of density of wire.

$$\frac{\Delta\rho}{\rho} \times 100 = \frac{\Delta M}{M} \times 100 + 2 \times \frac{\Delta r}{r} \times 100 + \frac{\Delta L}{L} \times 100$$

$$= \frac{0.002}{0.4} \times 100 + 2 \times \frac{0.001}{0.3} \times 100 + \frac{0.02}{5} \times 100$$

$$= 0.5 + 0.67 + 0.4 = 1.57\% \approx 1.6\%$$

35. *(a)* Fundamental frequency of an open pipe, $f_0 = \frac{v}{2l}$

and fundamental frequency of a closed pipe,

$$f_c = \frac{v}{4l}$$

$$\therefore \quad \frac{f_0}{f_c} = \frac{v/2l}{v/4l} = \frac{2}{1}$$

$$\therefore \quad f_0 : f_c = 2 : 1$$

36. *(c)* Focal length of convex lens,

$$f_1 = +f$$

Focal length of concave lens,

$$f_2 = -f$$

When both lenses are kept in contact, then equivalent focal length (F) is given by

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{f} + \frac{1}{-f} = 0 \Rightarrow F = \infty$$

37. *(a)* Time period of satellite,

$$T = \frac{2\pi r}{v_0}$$

$$\Rightarrow \quad T = \frac{2\pi(R_e + h)}{v_0} \qquad \ldots(i)$$

Orbital velocity, $v_0 = \sqrt{\frac{GM_e}{r}}$

$$\Rightarrow \quad v_0 = \sqrt{\frac{GM_e}{(R_e + h)}} \qquad \ldots(ii)$$

From Eqs. (i) and (ii), we get

$$T = \frac{2\pi(R_e + h)}{\sqrt{\frac{GM_e}{R_e + h}}}$$

$$T = 2\pi \sqrt{\frac{(R_e + h)^3}{GM_e}} \qquad \ldots(iii)$$

Since, $M_e = \frac{4}{3} \pi R_e^3 d$

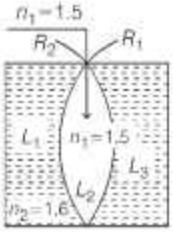
$\therefore$ Putting the value of M_e in Eq. (iii), we get

$$T = 2\pi \sqrt{\frac{(R_e + h)^3}{G \cdot \frac{4}{3} \pi R_e^3 d}} = \sqrt{\frac{3\pi(R_e + h)^3}{GdR_e^3}}$$

Since, satellite is orbiting just above the surface of the earth, hence $h = 0$

$$\therefore \quad T = \sqrt{\frac{3\pi R_e^3}{GdR_e^3}} = \sqrt{\frac{3\pi}{Gd}}$$

38. (*) The given situation is shown below.



$$\therefore \quad R_1 = 20 \text{ cm}$$

$$R_2 = -20 \text{ cm}$$

Lens L_1 and L_3 are plano concave lens.

Focal length of lens L_1 is given as

$$\frac{1}{f_1} = (n_2 - 1)\left(\frac{1}{\infty} - \frac{1}{R_1}\right)$$

$$= (1.6 - 1)\left(-\frac{1}{20}\right) = -\frac{1}{25}$$

$$\Rightarrow \quad f_1 = -25 \text{ cm}$$

Similarly, focal length of lens L_3,

$$f_3 = -25 \text{ cm}$$

Now, focal length of convex lens L_2 is given as

$$\frac{1}{f_2} = (n_1 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$= (1.5 - 1)\left(\frac{1}{20} - \frac{1}{-20}\right) = \frac{1}{20}$$

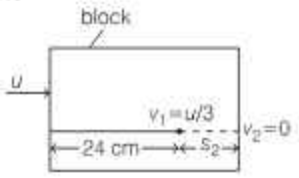
$$\Rightarrow \quad f_2 = 20 \text{ cm}$$

If F be the focal length of combination, then

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{-25} + \frac{1}{20} + \frac{1}{-25}$$

$$= -\frac{3}{100} \Rightarrow F = \frac{-100}{3} \text{ cm}$$

39. *(d)* The given situation is shown below.



$s_1 = 24$ cm

Using, $v^2 = u^2 - 2as$

$\Rightarrow \quad v_1^2 = u^2 - 2as \Rightarrow \left(\frac{u}{3}\right)^2 = u^2 - 2a \times 0.24$

$\Rightarrow 0.48a = \frac{8}{9}u^2 \Rightarrow a = \frac{8u^2}{9 \times 0.48} = \frac{u^2}{0.54}$

According to second condition,

$$0 = v_1^2 - 2as_2$$

$\Rightarrow \quad 0 = \left(\frac{u}{3}\right)^2 - 2 \times \frac{u^2}{0.54} \times s_2$

$\Rightarrow \quad \frac{u^2}{9} = \frac{2u^2}{0.54} s_2$

$\Rightarrow \quad s_2 = \frac{0.54}{9 \times 2} = 0.03 \text{ m} = 3 \text{ cm}$

$\therefore$ Total length of the block $= s_1 + s_2$

$= 24 + 3 = 27 \text{ cm}$

40. *(a)* When 10 resistors, each of resistance R is connected in series, then equivalent resistance.

$R_s = R + R + R + \ldots 10 \text{ times}$

$\Rightarrow \quad R_s = 10R$

$\therefore$ Current drawn from the battery,

$$I_s = \frac{E}{R_S} = \frac{E}{10R}$$

$\Rightarrow \quad I_s = \frac{E}{10R} \quad \ldots(i)$

In second case, when resistors are connected in parallel, then equivalent resistance,

$\frac{1}{R_p} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} + \ldots 10 \text{ times}$

$= \frac{10}{R}$

$\Rightarrow \quad R_p = \frac{R}{10}$

$\therefore$ Current drawn from the battery,

$I_p = \frac{E}{R_p} = \frac{E}{R/10} = 10 \times \frac{E}{R}$

$= 100 \times \frac{E}{10R} \Rightarrow I_p = 100I$

$\therefore \quad n = 100$

41. *(b)* Given, $L = \frac{50}{\pi} \text{ mH} = \frac{5}{\pi} \times 10^{-2} \text{ H}$

$C = \frac{10^3}{\pi} \mu\text{F} = \frac{1}{\pi} \times 10^{-3} \text{ F}$

$R = 10\,\Omega, f = 50 \text{ Hz}, V = 220 \text{ V}$

$\therefore \quad X_L = 2\pi fL$

$= 2\pi \times 50 \times \frac{5}{\pi} \times 10^{-2} = 5\,\Omega$

$X_C = \frac{1}{\omega C} = \frac{1}{2\pi fC}$

$= \frac{1}{2\pi \times 50 \times \frac{1}{\pi} \times 10^{-3}}$

$= \frac{1}{10^{-1}} = 10\,\Omega$

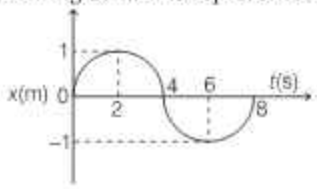
$\therefore$ Net impedance the circuit,

$Z = \sqrt{R^2 + (X_C - X_L)^2}$

$= \sqrt{10^2 + (10 - 5)^2}$

$= \sqrt{125} = 5\sqrt{5}\,\Omega$

42. *(c)* The motion of the particle performing S.H.M. is represented by



$x = A \sin \omega t = 1 \sin \frac{2\pi}{T} t = 1 \sin \frac{2\pi}{8} t$

$\Rightarrow \quad x = \sin\left(\frac{\pi}{4} t\right)$

$\therefore$ velocity, $v = \frac{dx}{dt}$

$= \frac{d}{dt} \sin\left(\frac{\pi}{4}\right) t \quad = \frac{\pi}{4} \cos\left(\frac{\pi}{4}\right) t$

$\therefore$ Acceleration of the particle,

$a = \frac{dv}{dt} = \frac{d}{dt}\left[\frac{\pi}{4} \cos\left(\frac{\pi}{4}\right) t\right]$

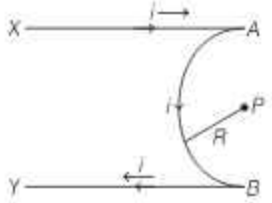
$= -\frac{\pi}{4} \cdot \frac{\pi}{4} \cdot \sin\left(\frac{\pi}{4} t\right)$

$\Rightarrow \quad a = -\frac{\pi^2}{16} \sin\left(\frac{\pi}{4} \cdot t\right)$

At $\quad t = 2 \text{ s}$

$a = -\frac{\pi^2}{16} \cdot \sin\left(\frac{\pi}{4} \cdot 2\right) = -\frac{\pi^2}{16} \text{ ms}^{-2}$

43. *(b)*



Net magnetic field at point P,

$B = B_{\text{semicircle}} - B_{XA} - B_{YB}$

$= \frac{\mu_0 i}{4R} - \frac{\mu_0 i}{4\pi R} - \frac{\mu_0 i}{4\pi R}$

$= \frac{\mu_0 i}{4R}\left[1 - \frac{2}{\pi}\right]$ pointed away from the page.

44. (*) Electric potential at point P due to electric dipole,

$V = \frac{1}{4\pi\varepsilon_0} \cdot \frac{p}{r^2} = K \cdot \frac{p}{r^2}$

$= K \cdot \frac{(q \cdot 2a)}{r^2}$

Here, $r = 5 \text{ cm} = 5 \times 10^{-3} \text{ m}$

$2a = (3 + 3) \text{ cm} = 6 \text{ cm}$

$= 6 \times 10^{-2} \text{ m}$

$\therefore \quad V = K \cdot q \cdot \frac{6 \times 10^{-2}}{(5 \times 10^{-2})^2}$

$= Kq \cdot \frac{6}{25} \times 10^2$

$= Kq\left(\frac{6}{25}\right) \times (10^2 \text{ V})$

$= \left(\frac{6}{25}\right) Kq \times (10^2 \text{ V})$

45. *(b)* Given, $R_0 = 2\,\Omega$

At $t = 80°\text{C}, R_t = 6.8\,\Omega$

We know that,

$R_f = R_0 (1 + \alpha t)$

$\Rightarrow \quad R_t = R_0 + R_0 \alpha t$

$\therefore \quad \alpha = \frac{R_t - R_0}{R_0 t}$

$= \frac{6.8 - 2}{2 \times 80} = \frac{4.8}{2 \times 80} = 0.03$

$= 3 \times 10^{-2} \,^\circ\text{C}^{-1}$

46. *(b)* Given, $\mathbf{L} = L\hat{\mathbf{i}}$

$\mathbf{B} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 4\hat{\mathbf{k}}$

$\therefore \quad \mathbf{F} = I(\mathbf{L} \times \mathbf{B})$

$= I\,[L\hat{\mathbf{i}} \times (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} - 4\hat{\mathbf{k}})]$

$= I\,[L(\hat{\mathbf{i}} \times \hat{\mathbf{i}}) + 3L\,(\hat{\mathbf{i}} \times \hat{\mathbf{j}}) - 4L\,(\hat{\mathbf{i}} \times \hat{\mathbf{k}})]$

$= I\,[0 + 3L\hat{\mathbf{k}} + 4L\hat{\mathbf{j}}]$

$= I\,[4L\hat{\mathbf{i}} + 3L\hat{\mathbf{k}}]$

$\therefore \quad F = |\mathbf{F}| = \sqrt{(4LI)^2 + (3LI)^2}$

$= 5LI$

47. *(b)* Given, $\mu = 0.15$

For the given situation,

$F = ma = \mu mg \Rightarrow a = \mu g$

$= 0.15 \times 10$

$= 1.5 \text{ ms}^{-2}$

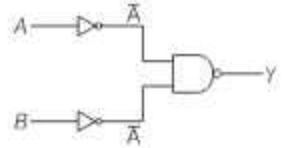
48. *(c)* Radius of innermost orbit of H-atom

$r_0 = 5.3 \times 10^{-11} \text{ m}$

$\therefore$ Radius of third allowed orbit of H-atom,

$$r_3 = r_0 n^2 = r_0 (3)^2 = 9r_0$$
$$= 9 \times 5.3 \times 10^{-11}$$
$$= 47.7 \times 10^{-11}$$
$$= 4.77 \times 10^{-10} \text{ m}$$
$$= 4.77 \text{ Å}$$

49. *(a)* Output of the logic circuit is



$$Y = \overline{\overline{A}\,\overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} \quad (\because \overline{xy} = \bar{x} + \bar{y})$$
$$= A + B \quad (\because \overline{\bar{x}} = x)$$
= output of OR gate

Truth table is

A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

50. *(b)* Maximum height attaine by the ball, $h' = \frac{v^2}{2g} = \frac{4^2}{2 \times 10} = 0.8$ m

Time of flight from bridge to bridge

$$t_1 = \frac{2v}{g} = \frac{2 \times 4}{10} = 0.8 \text{ s}$$

Total time of flight is given as 4 s.

Time taken by the ball to reach at the surface of water from highest point,

$$t_2 = 4 - \frac{t_1}{2} = 4 - \frac{0.8}{2} = 3.6 \text{ s}$$

$$\therefore \quad h = 0 \times t_2 + \frac{1}{2} g t_2^2$$
$$= 0 + \frac{1}{2} \times 10 \times (3.6)^2 = 64.8 \text{ m}$$

∴ The height of bridge above surface of water

$$= h - h'$$
$$= 64.8 - 0.8 = 64 \text{ m}$$

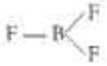
Chemistry

51. *(d)* We know that, $n_m = 2l + 1$.

$$\therefore \quad 2l = n_m - 1$$

and $$l = \frac{n_m - 1}{2}$$

52. *(b)* Among the given option BF_3 molecule acts as a Lewis acid. As it is an electron deficient molecule and has vacant orbital to accept the electron pair.



BF_3 has six electrons around central atom B that means its octet is not complete.

53. *(b)* Among the given statements, D and E are not correct because H—H bond dissociation enthalpy is not the lowest because bond formed between hydrogen atom is due to overlapping of 1s-1s orbital. So, it has smaller bond length and higher, bond enthalpy. There are also many other bond which are less stable then H—H and thus have less bond dissociation enthalpy.

Next, hydrogen does not reduces oxides of metals that are more active than iron. It can reduces oxides of elements like copper, mercury, silver, gold etc. as these metals are less reactive than hydrogen.

54. *(d)* Among the given options. Only option (d) is correct while other options are incorrect. The correct form of other statements are as follows.

(a) All enzymes that utilise ATP in phosphate transfer require Mg as the co-factor.

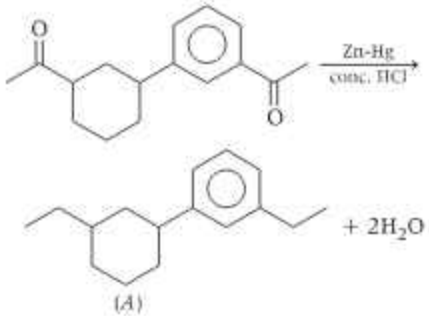
(b) The bone in human body is not inert and unchanging substance. It changes throughout the life.

(c) Ca plays roles in neuromuscular function and interneuronal transmission.

55. *(d)* Among the given option potassium trioxalatoaluminate (III) is an example of homoleptic complex while all other are examples of hetroleptic complex.

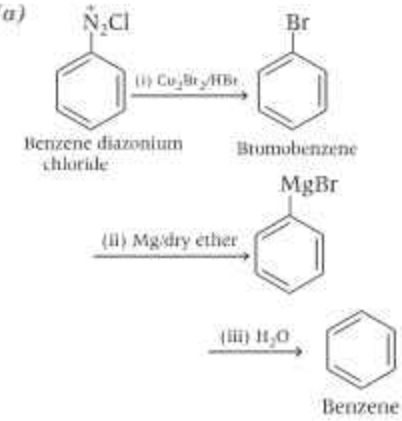
Complexes in which a metal is bound to only one kind of donor groups are known as **homoleptic complexes**. Complexes in which a metal is bonded to more than one kind of donor groups are known as hetreroleptic complexes.

56. *(d)* The final reaction is as follows.



This reaction is an example of Clemmensen reduction reaction.

57. *(a)*



58. *(d)* The correct order of energies of molecular orbitals of N_2 molecule is,

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s <$
$(\pi 2p_x = \pi 2p_y) < \sigma 2p_z$
$< (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$

N_2 has total of 14 electrons. Thus, electrons can be filled as

$\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2$
$< (\pi 2p_x^2 = \pi 2p_y^2) < \sigma 2p_z^2$

59. *(b)* Given, rate $(r) = k[A]^2[B]$ ...(i)

$$r_1 = k[3A]^2[B] \quad \text{...(ii)}$$

Dividing Eq (i) by (ii), we get

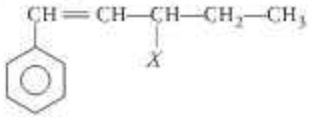
$$\frac{r}{r_1} = \frac{k[A]^2[B]}{k[3A]^2[B]}$$
$$\frac{r}{r_1} = \frac{1}{9}$$
$$\Rightarrow \quad r_1 = 9r$$

The initial rate would increase by a factor of nine.

60. *(d)* Both (A) and (R) are correct and (R) is the correct explanation of (A). The given relation i.e. $\Delta_r G = -nFE^\circ_{cell}$ is correct. Thus, according to this relation $\Delta_r G$ depends on n.

Also as ΔG depends on n i.e. number of mole, thus it is an extensive property and E_{cell} is independent of the amount of substance thus, it is an intensive property. As extensive properties are dependent on the amount of the substances.

61. *(b)* The given compound



- is an example of allylic halides. As in this the halogen is bonded to the sp^3-hybridised carbon atom next to the carbon-carbon double bond, which is called as allylic halides.
- **Benzylic halides** In these halides, the halogen atom is bonded to the sp^3-hybridised carbon atom next to an aromatic ring.
- **Vinylic halide** In these halides, the halogen atom is bonded to the sp^2-hybridised carbon of one of the carbon atoms of a double bond.
- **Aryl halide** In these halides, the halogen atom is bonded to the sp^2-hybridised carbon atom of an aromatic ring.

62. *(a)* Chloroprene $[CH_2=\underset{\displaystyle Cl}{\underset{|}{C}}-CH=CH_2]$ is the monomer of neoprene.

$$x\,CH_2=\underset{\displaystyle Cl}{\underset{|}{C}}-CH=CH_2 \xrightarrow{\text{Polymerisation}} \left[-CH_2-\underset{\displaystyle Cl}{\underset{|}{C}}=CH-CH_2-\right]_n$$

Chloroprene — Neoprene

63. *(d)*

$$CH_3\underset{\displaystyle CH_3}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3 \xrightarrow{H^+} CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-\overset{+}{C}H-CH_3$$

3-methylbutan-2-ol — 2° carbocation (Less stable)

$$\xrightarrow{\text{Hydride shift}} CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{+}{C}}}-CH_2-CH_3 \xrightarrow{Br^-}$$

3° carbocation (More stable)

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{Br}{\overset{|}{C}}}}-CH_2-CH_3$$

2-bromo-2-methyl butane

64. *(b)* Given, conductivity (κ) = 0.0210 $\Omega^{-1}cm^{-1}$

Resistance = 60 Ω

Cell constant = ?

We know that, $\kappa = \frac{1}{R}\times\frac{l}{a}$

Cell constant $= \frac{l}{a} = \kappa \times R = 0.0210 \times 60 = 1.26\ cm^{-1}$

65. *(b)* The correct match is A—III, B—I, C—IV, D—II.

66. *(b)*

Given elements	Na	N	O	F
Formed ions to attain by nearest nobel gas configuration	Na^+ (10 e^-)	N^{3-} (10 e^-)	O^{2-} (10 e^-)	F^- (10 e^-)

Thus, all are isoelectronic species. In case of isoelectronic species, more is the negative charge on the ion, lesser is the effective nuclear charge, more will be the radius. Hence, N^{3-} has the largest radius among the above ions.

67. *(b)* Among the given reactions, primary amine will not form in case of 2nd reaction.

- $CH_3NC \xrightarrow[H_2O]{LiAlH_4} CH_3-\underset{\displaystyle H}{\underset{|}{N}}-CH_3$ (2° amine)
- $CH_3-CN \xrightarrow[H_2O]{LiAlH_4} CH_3CH_2NH_2$ (Primary amine)
- $CH_3-\underset{\displaystyle O}{\underset{\|}{C}}-NH_2 \xrightarrow[H_2O]{LiAlH_4} CH_3CH_2NH_2$ (Primary amine)
- $CH_3CONH_2 \xrightarrow{Br_2/NaOH} CH_3NH_2$ (Primary amine)

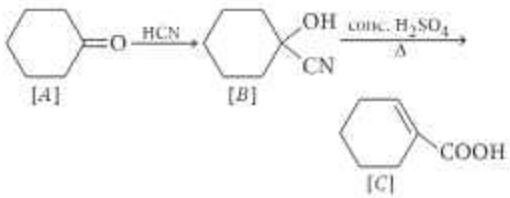
68. *(b)*



In pyridine the number of σ, π and lone pair are 11, 3 and 1 respectively.

69. *(b)* Statement I is correct while statement II is incorrect. The correct form of statement II is — When nucleoside is linked to phosphoric acid at 5′-position of sugar moiety, we get nucleoside.

70. *(c)*



[A] — [B] — [C]

71. *(c)* Among the given relation $TlI > TlI_3$ is correct. As in group 13 element +1 oxidation state becomes more and more stable as we go down the group i.e. Al < Ga < In < Tl. This is because of the presence of intervening *d* and *f*-orbitals which do not shield the nucleus efficiently. Consequently effective nuclear charge increases, that hold the *ns* electron tightly, so that they do not participate in bonding (inert pair effect). Thus, in Tl +1 oxidation state is more predominent whereas the +3 is strongly oxidising in nature.

72. *(c)* Among the given tranquilisers, veronal belongs to barbiturates.

73. *(b)* (A) is true while (R) is false. This is because at normal temperature and pressure helium is practically insoluble in oxygen. However under extreme conditions very small amount of helium dissolve in oxygen.

74. *(a)* Given reaction, $CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

Molar mass = 100 g/mol

20 g CO_2 of 20% pure limestone on heating gives $\frac{20\times 20}{100} = 4$ g

$$\text{Mole} = \frac{4}{100} = 0.004$$

Mass of CO_2 produces = Mole × Molecular mass of CO_2

$= 0.004 \times 44 = 1.76$ g

75. *(c)* If the compound contains sulphur and nitrogen both, sodium sulphocyanide is formed in Lassaigne's extract, which gives blood red colouration of $Fe(SCN)_3$ with $FeCl_3$.

$$Fe^{3+} + SCN^- \longrightarrow \underset{\text{Blood red}}{Fe[(SCN)]^{2+}}$$

76. *(c)* (A) is false but (R) is true. When a reaction have zero activation energy, the initial energy of the reactant will be equal to the activation energy of the reaction. That means every collision results into a chemical reaction which is not possible.

Mathematically,

If $E_a = 0$, the according to Arrhenius equation

$$k = Ae^{-E_a/RT} = Ae^0 = A$$

77. *(d)* The element B forms cubic close packed structure = 4

Atom A occupy $\frac{1}{3}$ tetrahedral voids $= \frac{1}{3} \times 8$

[$\because$ Tetrahedral voids = 2 of atom occupy by B]

$\therefore$ The formula of the compound is $A_{\frac{8}{3}}B_4$ or A_2B_3

$\therefore$ The value of $x + y = 2 + 3 = 5$.

78. *(b)* In aqueous solution, copper (I) undergoes disproportionation reaction.

$$2Cu^+(aq) \longrightarrow Cu^{2+} + Cu(s)$$

The highest stability of Cu^{2+} ion in aqueous solution is due to negative enthalpy of hydration.

79. *(c)* Among the given statements (B), (C) and (E) are correct while remaining are incorrect. The correct form of (A) and (D) are as follows.

(A) Atoms of all elements are composed of three fundamental particles i.e. electrons, protons and neutrons.

(B) Protons and neutrons are collectively known as nucleons.

80. *(a)* According to Boyle's law, at constant temperature, the pressure of a fixed amount of a gas varies inversely with its volume. $p \propto \frac{1}{V}$ or $p = (nRT)\frac{1}{V}$

Compare it with straight line equation, $y = mx + C$

we will get, $y = p$, $x = \frac{1}{V}$ with $C = 0$

and slope $(m) = T$

Thus, From the given options we can clearly see that, the correct option is given in option (a) i.e. p *vs* $1/V$ is a straight line with $T_3 > T_2 > T_1$.

81. *(b)* Among the given options covalent bonding is not an intermolecular force. It is the bond which is formed by mutual sharing of electrons between the combining atoms of the same or different element. Whereas dipole-dipole forces, dipole-induced dipole forces, hydrogen bonds and dispersion forces are intermolecular forces.

82. *(a)* $CH_3-\underset{\underset{\text{Sodium ethanoate}}{\underset{\|}{O}}}{\overset{}{C}}-\overset{-}{O}\overset{+}{Na} + NaOH \xrightarrow[\text{Decarboxylation}]{CaO,\ \Delta} CH_4 + Na_2CO_3$

The product formed in above reaction is CH_4. Thus, weight of two moles of CH_4 is 2×12 i.e. 32.

83. *(c)* Among the given cases, option (c) is an example of heterogeneous catalysis as in this case the reactant and the catalyst are present in different phase.

Whereas all other are example of homogeneous catalysis as in respective reaction the reactants and the catalyst are present in same phase.

(a) $\underset{\text{Sugar solution}}{C_{12}H_{22}O_{11}(aq)} + H_2O(l) \xrightarrow{H_2SO_4(l)} \underset{\text{Glucose}}{C_6H_{12}O_6(aq)} + \underset{\text{Fructose}}{C_6H_{12}O_6(aq)}$

(b) $NO(g) + O_3(g) \longrightarrow O_2(g) + NO_2(g)$

$NO_2(g) + O(g) \longrightarrow NO(g) + O_2(g)$

[These reactions are involve in ozone depletion through nitrous oxide.]

(c) $N_2(g) + 3H_2(g) \xrightarrow{Fe(s),\ Mo(s)} 2NH_3(g)$

(d) $2SO_2(g) + O_2(g) \xrightarrow{NO(g)} 2SO_3(g)$

84. *(b)* (A) is true but (R) is false.

(A) is correct because all alkali metals gives deep blue solution by giving electrons.

When sodium metal dissolved in liquid ammonia following reaction takes place.

$$Na + (x + y)\,NH_3 \longrightarrow \underset{\text{Ammoniated cations}}{[Na(NH_3)_x]^+} + \underset{\text{Ammoniated electrons}}{2[e^-(NH_3)_y]^-}$$

Thus, the correct (R) statement is:

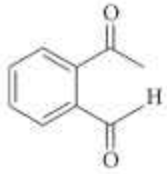
The ammoniated electrons are responsible for the blue colour of the solution as they absorb the energy from visible region of light and, thus impart blue colour to the solution.

85. *(d)*

Given species	NH_3	$AlCl_3$	$BeCl_2$	CCl_4	PCl_5
Number of electrons central atom	8	6	4	8	10

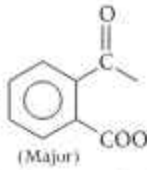
Thus, among the given species $AlCl_3$, $BeCl_2$, PCl_5 does not have eight electrons around the central atom.

86. *(b)* The complete given reaction is as follows.



$+ 2[Ag(NH_3)_2] + 3\,\bar{O}H \xrightarrow[H_2O]{\Delta}$

[Tollen's reagent]



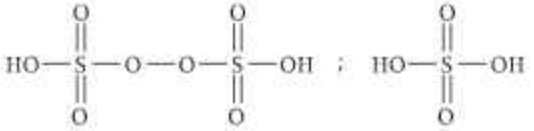
$H_2O + NH_3 +$ $+ Ag\downarrow$ Silver

(Major)

Tollen's reagent oxidises only the aldehyde group in the above reaction not the ketonic group.

87. *(a)* The correct match is A—III, B—IV, C—I, D—II.

Structure of given oxoacids of sulphur are as follows

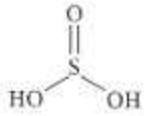


Peroxodisulphuric acid
[Two S—OH,
four S=O
one S—O—S bonds.]

Sulphuric acid
[Two S—OH,
two S=O bonds.]

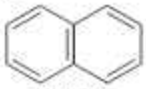


Pyrosulphuric - acid
[Two S—OH,
four S=O bond,
one S—O—S bonds]



Sulphrous acid
[Two S—OH,
one S=O bond.]

88. *(d)* Among the given species that follows Huckel rule are given as



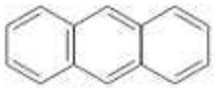
Benzene
[No. of $\pi e^-=6$]



Cyclopentadienyl
[No. of $\pi e^-=6$]



Cyclopropenyl cation
[No. of $\pi e^-=2$]



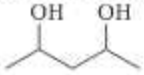
Anthracene
[No. of $\pi e^-=14$]

Whereas all other are non-aromatic and does not follows Huckel rule, as (iii), (iv) and (vi) have 4, 4 and 8 π electrons respectively.

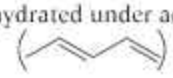
89. *(b)* Pumice stone is an example of solid sol. In this dispersion phase is gas and dispersion medium is solid.

90. *(a)* Among the given option the correct relation between change in enthalpy (ΔH) and change in internal energy (ΔU) is, $\Delta H = \Delta U + \Delta n_g RT$

91. *(a)* Among the given alcohols,

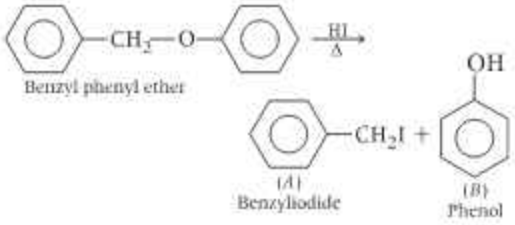


will most readily dehydrated under acidic conditions to give



1, 3 butadiene as a product, in which double bonds are present in conjugation.

92. *(b)* Benzyl phenyl ether on heating with HI gives benzyl iodide (A) and phenol (B). The complete reaction is as follows.



Benzyl phenyl ether

Benzyliodide (A)

Phenol (B)

93. *(d)* The value of a, b and c is 1, 3, 8 the complete balanced equation is as follows.

$$Cr_2O_7^{2-} + 3SO_3^{2-}(aq) + 8H^+(aq) \longrightarrow$$

$$2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + \frac{8}{2}H_2O(l)$$

94. *(b)* Among the given complex $[Co(Cl_2)(en)_2]NO_3$ is most stable because it is a chelating complex.

It coordinates with central metal atom to form a ring compound. Due to this reason, there is more force of attraction between the central metal ion and the chelating agent therefore it is more stable.

95. *(b)* As we know that, octahedral voids are present at the centre of edge length of the cube. Also, if any atom is present at the edge length then it divide into 1 / 4 parts. Thus, fraction of one edge centred octahedral void lies in one unit cell of *fcc* is 1 / 4.

96. *(b)* Among the given statements C and D statements are incorrect. Their correct forms are as follows.

(C) Basic character decreases from V_2O_3 to V_2O_4 to V_2O_5. In general, the oxides in lower oxidation states of metals are basic and in their higher oxidation states they are acidic.

(D) V_2O_4 dissolves in acids to give VO_2^+ salts.

97. *(b)* According to given, general equation

$$A + B \rightleftharpoons C + D$$

$$\text{Equilibrium constant } (K_{eq}) = \frac{(10)(6)}{(2)(3)} = \frac{60}{6}$$

$$K_{eq} = 10$$

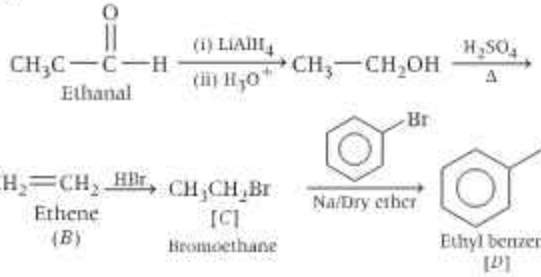
We know that,

$$\Delta G^\circ = -2.303\ RT \log K_{eq}$$

$$\therefore \quad \Delta G^\circ = -2.303 \times 2 \times 300 \times \log(10)$$

$$= -1381.8\ \text{cal}$$

98. *(d)*



Ethanal

Ethene (B)

Bromoethane [C]

Ethyl benzene [D]

99. *(e)* Statement I is false but statement II is true. The correct statement I is : The nutrient rich water bodies lead to eutrophication.

The process in which nutrient enriched water bodies support a dense population, which kills animals life by depriving it of oxygen and results in subsequent loss of biodiversity is known as eutrophication.

100. *(d)* Among the given reaction,

$Fe_2O_3 + CO \longrightarrow 2FeO + CO_2$ reaction does not occur between 900 K to 1500 K temperature range in the blast furnace during extraction of iron. This reaction occur in temperature range of about 500-800 K (i.e. reduction zone).

Botany

101. *(a)* Among eukaryotes, replication of DNA occurs during the synthetic or S-phase of the cell cycle.During this phase, the cell synthesises an exact copy of its DNA. After DNA replication is complete, the cell proceeds to the G_2-phase, during which it checks the replicated DNA for errors and prepares for mitosis (M-phase), i.e. the process of cell division.

102. *(b)* Cellulose does not form blue colour with iodine solution because it does not contain the helical structure which is present in amylose. Amylose forms an iodine-amylose complex with iodine, which results in a blue-black colour. If iodine molecules come into contact with cellulose, they cannot penetrate the tightly packed and rigid structure of cellulose chains to form a complex.

103. *(b)* In gene gun method, used to introduce alien DNA into host cells, microparticles of tungsten or gold are used. These metals are dense, inert and biologically non-reactive metals that can penetrate the cell wall and enter the cell. In this method, small microparticles made of tungsten or gold are coated with the DNA of interest and then accelerated to high velocities using a device called a gene gun. This technique is also called biolistic method.

104. *(a)* In a corn cob, tassels are used to trap pollen grains. The tassels on a corn cob are the stigma and style which wave in the wind to trap pollen grains.

105. *(d)* Both A and R are true and R is the correct explanation of A. Latewood or summerwood is the part of a tree's growth ring that is produced later in the growing season. It has fewer xylary elements and narrower vessels than the earlywood. The latter is produced earlier in the growing season.The cambium is the layer of cells between the bark and wood of a tree that is responsible for producing new xylem and phloem cells. During the growing season, the cambium produces a large number of new xylem cells which result in the formation of wide vessels in the earlywood.Later in the growing season, as the cambium slows down its production of xylem cells, the vessels become narrower and fewer in number, resulting in the formation of latewood.

106. *(a)* The historic Convention on Biological Diversity "The Earth Summit" was held in Rio de Janeiro, Brazil in the year 1992.The summit was organised by the United Nations and was attended by representatives from over 170 countries. The goal of the Conventionwas to promote the conservation of biodiversity, the sustainable use of its components, etc.

107. *(a)* The role of RNA polymerase-III (Pol-III) in eukaryotes is the transcription of several types of small non-coding RNAs, including transfer RNAs (tRNAs), 5S ribosomal RNA (5S rRNA), and some small nuclear RNAs (snRNAs).tRNAs are involved in protein synthesis, 5S rRNA is a component of the ribosome, and snRNAs play a role in RNA splicing. Pol-III binds to specific DNA sequences called promoters and synthesises RNA using the DNA strand as a template.

108. *(a)* Among the given options, *Selaginella* and *Salvinia* represent the pair of heterosporous pteridophytes.Heterosporous pteridophytes are plants that produce two types of spores namely, megaspores and microspores. These spores develop into female and male gametophytes, respectively.

109. *(d)* Both Statement I and Statement II are correct. Transpiration is the process by which plants lose water vapour through their leaves. This loss of water creates a negative pressure in the xylem vessels that extends from the roots to the leaves. It in turn generates a force known as the transpiration pull, that can lift water upward over 130 meters in height.Additionally, transpiration can have a cooling effect on the leaf surface due to evaporative cooling. As water is lost through the stomata on the leaf surface, heat is absorbed from the surrounding tissues and the leaf surface can be cooled by as much as 10-15 degrees Celsius.

110. *(d)* The reaction centre in photosystem-II (PS-II) has an absorption peak at around 680 nanometers (nm) in the visible light spectrum. PS-II is one of the two photosystems found in the thylakoid membranes of chloroplasts and is responsible for generating the oxygen released into the atmosphere as a by-product of photosynthesis. The other photosystem, photosystem-I (PS-I), has an absorption peak at around 700 nm and is responsible for producing NADPH.

111. *(d)* Among "The Evil Quartet", habitat loss and fragmentation are the most important cause of driving extinction of species. As habitats are destroyed or broken up into smaller and more isolated fragments, species become more vulnerable to other threats such as predation, disease, and climate change. Habitat loss and fragmentation can have direct and indirect impacts on species, including loss of resources, increased isolation, and reduced genetic diversity.

112. *(b)* The division of the centromere occurs during the anaphase-II stage of meiosis. During this stage, the sister chromatids of each chromosome separate and are pulled toward opposite poles of the cell by spindle fibers. The separation of the sister chromatids is facilitated by the division of the centromere, which is the specialised region of the chromosome that links the sister chromatids together.

113. *(a)* The spraying of Gibberellic Acid (GA) on juvenile conifers can help in hastening the maturity period, leading to early seed production. Gibberellic acid is a plant hormone that plays a key role in promoting stem and leaf growth, seed germination, and flowering. In conifers, GA can help to promote growth and hasten the maturity period thus, leading to early seed production.

114. *(d)* Among the given statements, A, B and C are correct whereas statements D and E are incorrect. The incorrect statements can be as the detritus food chain begins with the breakdown of dead organic matter, also known as detritus. It includes dead plant and animal material, as well as waste products like faeces and fallen leaves. The process by which earthworms breakdown the complex detritus into smaller particles is called fragmentation.

115. *(d)* The micronutrient required for splitting water molecules during photosynthesis is manganese (Mn). In the light-dependent reactions of photosynthesis, light energy is used in the photolysis of water molecules into oxygen, protons, and electrons. This process occurs in the thylakoid membranes of chloroplasts and is facilitated by a complex of proteins called Photosystem-II (PS-II). Manganese is a critical cofactor for the PS-II complex, as it plays a key role in binding and stabilising the water molecules during the photolysis reaction.

116. *(d)* Large, colourful, fragrant flowers with nectar are often seen in plants that are pollinated by animals, particularly insects. These flowers have evolved to attract pollinators, which transfer pollen from the anthers of one flower to the stigma of another flower of the same species. Some examples of plants that produce large, colourful, fragrant flowers with nectar include roses, lilies, *Hibiscus*, and magnolias.

117. *(c)* The movement and accumulation of ions across a membrane against their concentration gradient can be explained by active transport. It is a process by which ions or molecules are moved across a cell membrane against their concentration gradient, from a region of lower concentration to a region of higher concentration. This requires energy in the form of ATP and the use of specific transport proteins or pumps.

118. *(d)* The thickness of ozone in a column of air in the atmosphere is measured in the terms of Dobson unit. This unit is commonly used by atmospheric scientists to monitor changes in the thickness and distribution of the ozone layer over time.

119. *(c)* Upon exposure to UV radiation, DNA stained with ethidium bromide will show bright orange colour. Ethidium bromide is a fluorescent dye that binds to DNA and becomes brightly coloured under ultraviolet (UV) light. When exposed to UV radiation, the dye intercalates into the double helix structure of DNA and emits a bright orange-red colour. This property is often used in molecular biology and genetics research to visualise and analyse DNA fragments in agarose gel.

120. *(a)* Unequivocal proof that DNA is the genetic material was first proposed by Alfred Hershey and Martha Chase. In 1952, Alfred Hershey and Martha Chase conducted a series of experiments using a virus called bacteriophage T_2 to infect bacteria. They demonstrated that only the DNA of the virus, and not its protein coat, was transferred into the bacterial cell. This experiment, known as the Hershey-Chase experiment, provided strong evidence that DNA, and not protein, was the genetic materials responsible for passing on the traits from one generation to the next.

121. *(a)* In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form a callus. This phenomenon may be called dedifferentiation. It refers to the process by which mature cells lose their particular characteristics and they closely resemble embryonic cells. In tissue culture experiments, leaf mesophyll cells can be induced to undergo dedifferentiation and form callus when placed in a culture medium that contains the right combination of nutrients, hormones, and growth factors. The cells in the callus can then be induced to differentiate and form new tissues or organs.

122. *(d)* Both A and R are true and R is the correct explanation of A.

In glycolysis, glucose is converted into pyruvate through a series of enzymatic reactions, and ATP is used in two steps. First, it is used when glucose is phosphorylated to glucose-6-phosphate by hexokinase, using one ATP molecule. Second, when fructose-6-phosphate is phosphorylated to fructose-1,6-bisphosphate by phosphofructokinase, using another ATP molecule.

123. *(a)* The appearance of recombination nodules occurs during the pachytene substage of prophase I in meiosis. During this stage, the homologous chromosomes continue to undergo synapsis and become fully paired, forming structures known as bivalents or tetrads. At this point, the recombination nodules become more visible along the paired chromosomes.

124. *(b)* In the given equation, R represents respiratory loss. The rate at which organic matter is produced during photosynthesis is referred to as the ecosystem's Gross Primary Productivity (GPP). When we subtract respiratory losses (R) from GPP, we get Net Primary Productivity (NPP). The available biomass for heterotrophic consumption is defined as net primary productivity.

125. *(b)* Pleiotropism refers to a phenomenon where a single gene can affect multiple traits or phenotypic expressions. In other words, a single gene has multiple effects on the organism's phenotype. For example, in phenylketonuria, a single mutation in the gene responsible for the production of an enzyme causes hypopigmentation of the hair and skin, as well as mental retardation.

126. *(a)* In *r*DNA technology, the addition of chilled ethanol precipitates out DNA. The chilled ethanol causes the DNA molecules to lose their solubility by disrupting the hydrogen bonds between the bases and the water molecules. The precipitated DNA can then be separated from the mixture.

127. *(b)* In angiosperms, the haploid structure is synergids, the diploid structure is zygote and the triploid structure is the primary endosperm nucleus. The synergids are two small, haploid cells that are present adjacent to the egg cell in the female gametophyte of the angiosperm. The diploid structure is the zygote, which is formed when the sperm cell fuses with the egg cell during fertilisation. The triploid structure is the endosperm, which is a tissue that nourishes the developing embryo.

128. *(c)* Axile placentation is observed in china rose, *Petunia*, and lemon. Axile placentation is a type of placentation where the ovules are attached to the central axis or the centre of the ovary, and the septa (walls) of the ovary divide it into two or more locules.

129. *(b)* In deep water rice, Ethylene promotes internode or petiole elongation. Ethylene is a plant hormone that regulates various physiological processes, including growth and development. In deep water rice, it promotes the elongation of internodes and petioles, which helps the plant to grow taller and maintain its leaves above the water surface to access air and sunlight.

130. *(d)* Both A and R are correct and R is the correct explanation of A.

The first stage of the gametophyte generation of mass is the protonema, which is a thread-like structure. It develops directly from the spores that are produced in the capsule of the sporophyte through meiosis.

The spores are released from the capsule and they germinate and grow into a protonema. The protonema is usually green and photosynthetic, and it may develop into a thalloid or leafy gametophore depending on the species of moss. The gametophore is the mature gametophyte plant and produces the gametes (sperm and eggs).

131. *(b)* Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on the chromosome was used for the first time by Alfred Sturtevant. His work paved the way for the development of genetic mapping techniques based on recombination frequencies, which are still widely used in modern genetics and genomics.

132. *(d)* Diadelphous and dithecous anthers are characteristics of the family Fabaceae, which includes many species of legumes such as beans, peas, and lentils. These anthers are composed of two fused pairs of pollen sacs, with one pair shorter than the other. In contrast, the families Solanaceae and Liliaceae have different types of anther arrangements. For example, in Solanaceae, anthers open by small pores to release the pollen, while Liliaceae have basifixed or dorsifixed anthers, meaning they are attached at the base or back of the anther.

133. *(a)* The synthesis of one molecule of glucose during Calvin cycle requires 18ATP and 12 $NADPH_2$.The Calvin cycle is a series of biochemical reactions that takes place in the stroma of chloroplasts during photosynthesis, where atmospheric CO_2 is converted into glucose. The synthesis of one molecule of glucose requires energy in the form of ATP and reducing power in the form of $NADPH_2$.

134. *(d)* Expressed Sequence Tags (ESTs) refer to all the genes that are expressed as RNA. ESTs are short DNA sequences that are generated by sequencing the transcribed regions of genes, specifically the complementary DNA (cDNA) produced from mRNA molecules. ESTs represent the expressed portion of a genome.

135. *(c)* Statement I is incorrect but statement II is true. Endarch, exarch, and mesarch are primary xylem while secondary xylem refers to the formation that occurs after the vascular cambium's secondary growth. It consists of larger-sized vessels and tracheids. Exarch is a characteristic feature of the root system of plants. In the roots of most plants, the protoxylem (the first-formed xylem) develops towards the periphery of the root, with the metaxylem (the later-formed xylem) located towards the centre of the root. This arrangement is called exarch.

136. *(d)* Chemiosmosis requires a membrane, proton pump, proton gradient and ATP synthase enzyme. During cellular respiration and photosynthesis, a proton pump actively transports protons (H^+) through proton pump across a selectively permeable membrane. It leads to the generation of a proton gradient. This gradient is used by the ATP synthase enzyme to produce ATP through the process of chemiosmosis.

137. *(d)* Ribosome consists of 80 proteins. The ribosomal proteins can be divided into two groups those present in the small ribosomal subunit and those present in the large subunit. The small ribosomal subunit has around 35 proteins whereas the large subunit has 53 proteins. In total, the ribosome consists of about 80 proteins, which help to stabilise the structure of the ribosome and facilitate its function in protein synthesis.

138. *(d)* Both A and R are correct and R is the correct explanation of A.

A flower is a modified shoot in which the shoot apical meristem changes to a floral meristem. The floral meristem produces floral appendages such as sepals, petals, stamens, and carpels. The internode of the shoot gets condensed to produce different floral appendages laterally at successive nodes instead of leaves. The floral appendages are arranged in a whorl-like pattern, and their number and arrangement vary among different plant species.

139. *(b)* A-IV, B-I, C-II, D-III.

M- phase is equational division as a parent cell divides to give rise to two equal- sized daughter cells. G_2-phase is characterised by the synthesis of proteins required for mitotic division. The quiescent phase is an inactive phase of the cell cycle wherein cells remain metabolically active and the G_1-phase represents an interval between mitosis and initiation of DNA replication.

140. *(d)* The correct sequence of the main steps in the formation of Recombinant DNA is B, C, D, A. First, the DNA is cut at specific locations using restriction enzymes in order to produce fragments of the desired size. The desired DNA fragment is then isolated and amplified using PCR. The recombinant DNA is then inserted into the host cell using various techniques such as transformation or electroporation. The insertion of recombinant DNA into the host cell results in the production of a genetically modified organism (GMO).

141. *(b)* Statement I is correct but statement II is false. The effects of competition on different organisms depend on several factors such as the type of resources being competed for, the availability of those resources, and the adaptations of the competing species. In some cases, carnivores may have an advantage over herbivores in competition because they are at a higher trophic level and require fewer resources to survive.

142. *(d)* A-II, B- IV, C- I, D- III

Cohesion represents mutual attraction among the water molecules and

adhesion is attraction towards polar surfaces. These forces are responsible for capillary action, which allows water to move up a narrow tube, such as the xylem vessel in plants. Surface tension is due to more attraction in the liquid phase. It is the measure of the cohesive forces between the molecules at the surface of a liquid, which creates a surface layer that resists external forces. Guttation is loss of water from the margins of the leaf in the form of liquid droplets.

143. *(d)* Melonate is a competitive inhibitor of succinate dehydrogenase, which is a key enzyme in the tricarboxylic acid (TCA) cycle or Krebs cycle. TCA cycle plays an essential role in the cellular respiration process of aerobic organisms, including bacteria. By inhibiting the activity of succinate dehydrogenase, melonate can disrupt the energy metabolism of pathogenic bacteria, leading to their growth inhibition.

144. *(c)* A- II, B- IV, C-III, D- I

Oxidative decarboxylation involves the conversion of pyruvate into acetyl-CoA. Pyruvate is first decarboxylated by pyruvate dehydrogenase, releasing carbon dioxide, and then oxidized to form acetyl-CoA.Glycolysis is also known as the Embden-Meyerhof-Parnas (EMP) pathway. Oxidative photophosphorylation occurs during the electron transport chain. In this process, energy is produced by the transfer of electrons from NADH and $FADH_2$ to oxygen. Tricarboxylic acid cycle requires citrate synthase, which catalyses the formation of citrate from acetyl-CoA and oxaloacetate.

145. *(b)* Statements B and E are correct whereas A, C, D are incorrect. The incorrect statements can be corrected as

- Klinefelter's Syndrome was first described by Harry Klinefelter in 1942, not by Langdon Down.
- It is not typically associated with short stature.
- Physical, psychomotor, and mental development may be affected in individuals with Klinefelter's Syndrome, but this can vary widely and is not necessarily always present. It is a usual symptom of Down's syndrome.

146. *(a)* Algal blooms caused by excess organic matter in water can lead to eutrophication, which can reduce water quality by depleting oxygen levels and producing toxins harmful to aquatic organisms, including fish. Eutrophication can also cause the loss of biodiversity and alter the ecological balance of the water body. So, algal blooms caused by excess organic matter do not improve water quality and promote fisheries.

147. *(b)* A is true but R is false.

In gymnosperms, the pollen grains are produced in the microsporangia, which are located in the male cones. The pollen grains are released from the microsporangium and are carried by air currents to the female cones, where they may land on the ovule. However, the male gametes are not discharged at the mouth of the archegonia. Instead, the pollen grain germinates on the surface of the ovule, and a pollen tube grows into the ovule to deliver the male gametes. The male gametes are then released from the pollen tube to fertilise the female gamete in the ovule.

148. *(b)* A- III, B- I, C- IV, D- II

Iron is an activator of catalase, which is involved in the breakdown of hydrogen peroxide, a toxic byproduct of cellular metabolism. Zinc is required for the activity of enzymes involved in auxin biosynthesis, and its deficiency can result in reduced auxin levels and stunted growth. Boron is essential for cell elongation and differentiation, which are key processes in plant growth and development. Molybdenum is an essential component of the enzyme nitrate reductase, which is involved in the reduction of nitrate to nitrite during nitrogen assimilation in plants.

149. *(a)* A- IV, B- I, C- II, D- III

In mutualism, both species A and B are benefited. In commensalism, species A is benefited whereas species B does not get any benefit. In amensalism, species A is negatively impacted whereas the other species B do not get any benefit. In parasitism, species A is benefitted whereas the species B is negatively impacted.

150. *(a)* Statements A and D are correct whereas B, C, and E are incorrect. These can be corrected as

- Bark formed early in the season is called soft bark.
- Bark is the non- technical term that reflects all tissues lying outside the vascular cambium.
- Phellogen is multi- layered in thickness.

151. *(a)* The ileo-cecal valve is a sphincter that separates the ileum from the caecum. The valve prevents the backflow of the contents of the colon into the small intestine.Once the undigested and unabsorbed substances enter the caecum, they are further acted upon by bacteria present in the colon. The colon absorbs water and electrolytes from the undigested material and moves it forward for elimination. The ileocecal valve ensures that this material moves forward in the digestive tract and is eventually eliminated from the body.

152. *(d)* The HIV (Human Immunodeficiency Virus) undergoes replication and produces progeny viruses in the T helper cells. HIV targets the immune system's T helper cells, which are a type of white blood cell that play a critical role in the body's defence against infection. The virus enters the T cells and takes over their machinery to replicate itself, leading to the destruction of the T cells.

153. *(d)* A broad palm with a single palm crease is a common physical feature seen in individuals with Down's syndrome. It is a genetic disorder caused by the presence of an extra copy of chromosome 21. It is characterised by a range of physical and intellectual disabilities. One of the common physical features seen in individuals with Down's syndrome is a broad palm with a single palmar crease. Normally, most people have two palmar creases on their palms, but in individuals with Down's syndrome, there is only one crease. This feature is also known as a "simian creaseor line".

154. *(a)* Gonorrhoea is a common sexually transmitted disease that is completely curable when detected early and treated properly. It is caused by the

bacterium *Neisseria gonorrhoeae* and can infect both men and women. The infection is transmitted through sexual contact with an infected partner. Symptoms can include painful urination, discharge from the penis or vagina, and pain or swelling in the testicles. If detected early, gonorrhoea can be easily treated with antibiotics.

155. *(d)* A-I, B-II, C-III, D-IV.

Leopard and lion in the same forest or grassland exhibit competition as they are both large carnivores occupying similar niches. Cuckoo laying eggs in a crow nest is an example of brood parasitism as it lays its eggs in the nest of another bird species for them to raise the cuckoo chicks. Fungi and roots of higher plants in mycorrhiza is an example of mutualism as both organisms benefit from the association.Cattle egrets and cattle in a field show commensalism as the cattle egret benefits by feeding on insects found on the cattle, while the cattle are unaffected.

156. *(b)* A- III, B- I, C-IV, D-II

Peptic cells are located in the gastric glands of the stomach and secrete pepsinogen, a precursor to the enzyme pepsin which helps in the breakdown of proteins. Goblet cells are also present in the stomach lining and produce mucus, which helps to protect the stomach wall from the corrosive action of gastric acid. Oxyntic cells are found in the lining of the stomach and secrete hydrochloric acid (HCl) and intrinsic factor, which is required for the absorption of vitamin-B_{12}. Hepatic cells are found in the liver and produce bile juice, which is stored in the gall bladder and released into the small intestine to help in the digestion of fats.

157. *(c)* Statement I is false but statement II is true. The N-terminal is the first amino acid in the protein sequence, and the C-terminal is the last amino acid. So, the left end of the protein would be the N-terminal and the right end would be the C-terminal.

158. *(d)* Both statement I and statement II are correct. RNA mutates at a faster rate because RNA polymerases lack proofreading ability and RNA viruses have high error rates during replication.

Also, RNA viruses have shorter generation times, larger population sizes, and higher mutation rates as compared to DNA viruses.
It results in rapid evolution and adaptation in RNA viruses to changing environments.

159. *(a)* A- III, B- IV, C- II, D- I

Vasectomy is the surgical method of contraception in males. Coitus interruptus is the natural method of contraception in which the penis is withdrawn just before the ejaculation. Cervical caps are used as the barrier method of contraception by females. Saheli is an oral non- hormonal contraceptive pill.

160. *(d)* A- II, B- III, C- IV, D- I

Ringworm is caused by fungi of the genus *Trichophyton*, filariasis is caused by the parasite *Wuchereria bancrofti*, malaria is caused by parasites of the genus *Plasmodium vivax*, and pneumonia can be caused by bacteria such as *Haemophilus influenzae*.

161. *(d)* A-IV, B- III, C- II, D- I

CCK (chole cystokinin) acts on both gall bladder and pancras and stimulates the secretion of Pancreatic and bile juice respectively.

GIP (Gastric Inhibitory Peptide) is a hormone secreted by the gastric gland that inhibits gastric secretion and motility. ANF (Atrial Natriuretic Factor) is a hormone secreted by the heart that regulates blood pressure and fluid balance. ADH (Antidiuretic Hormone) is a hormone secreted by the kidney that regulates water balance in the body.

162. *(d)* Both statement I and statement II are true.Vas deferens is a muscular duct that receives sperm from the epididymis and also receives a duct from seminal vesicle. The vas deferens opens into the urethra as the ejaculatory duct.The cervical canal is the cavity of the cervix that connects the uterus to the vagina. The vagina and the cervical canal together form the birth canal.

163. *(a)* Early diagnosis and understanding of a disease's pathogenesis are critical for efficient treatment.Early detection is not achievable by traditional diagnostic approaches (e.g. serum and urine analysis).Some of the procedures used for early diagnosis include recombinant DNA technology, Polymerase Chain Reaction (PCR), and Enzyme Linked Immunosorbent Assay (ELISA).

164. *(a)* A, C and E only is the correct answer as Mitochondria, chloroplasts and peroxisomes are not considered part of the endomembrane system while endoplasmic reticulum and Golgi complex are part of the endomembrane system.

165. *(d)* A- II, B- I, C- IV, D- III

Heroin is an opioid that slows down the central nervous system and can cause respiratory depression. Marijuana can affect the cardiovascular system by increasing heart rate and blood pressure. Cocaine interferes with the transport of dopamine, a neurotransmitter associated with pleasure and reward, and can lead to addiction. Morphine is a powerful painkiller that works by binding to opioid receptors in the brain and spinal cord.

166. *(d)* Both statement I and statement II are true. Low temperature slows down the enzyme activity and high temperature denatures the enzyme. Competitive inhibitor closely resembles the substrate and competes with it to bind to the active site of the enzyme, reducing the enzyme's activity.

167. *(b)* The cytoskeleton is a complex network of protein filaments that provides structural support, helps maintain cell shape, and facilitates various cellular functions. One of the key functions of the cytoskeleton is to enable cell movement or motility, both within the cell and through the environment. It is achieved through the interaction of its different components, including microfilaments, intermediate filaments, and microtubules, which work together to generate and transmit forces that produce movement.

168. *(a)* Both statement I and statement II are false. Ligaments are dense regular connective tissue that connects bones to other bones or cartilages. Cartilage is a type of supportive connective tissue that is avascular and

has no nerves. It is not classified as dense regular or dense irregular tissue.

169. *(a)* A- II, B- IV, C- I, D- III

- Cartilaginous joints are present between the adjacent vertebrae in the vertebral column. They allow for limited movement and provide support to the spine.
- Ball and socket joint is present between humerus and pectoral girdle. This type of joint allows for a wide range of movements in multiple planes, such as rotation.
- Fibrous joint is present between the flat skull bones. These joints are immovable and provide stability and protection to the brain.
- Saddle joint is present between carpal and metacarpal of thumb. This type of joint allows for a wide range of movements, including flexion, extension, abduction, adduction, etc.

170. *(a)* A- II, B- III, C- IV, D- I

Gene 'a' is associated with the function of transacetylase, which transfers an acetyl group from acetyl-CoA to various compounds, including lactose.

Gene 'y' is associated with the function of permease, which is a membrane protein that facilitates the transport of lactose into the cell. Gene 'I' is associated with the function of repressor protein, which binds to the operator site of the lac operon to prevent transcription of the genes involved in lactose metabolism.

Gene 'z' is associated with the function of β-galactosidase, which is an enzyme that breaks down lactose into glucose and galactose.

171. *(a)* Among the given options, (a) is correct.



This symbol shows consanguineous mating (marriage of blood relatives).

172. *(a)* Both A and R are true but R is NOT the correct explanation of A.

Endometrium is necessary for the implantation of blastocyst. It is the inner lining of the uterus that prepares every month to receive a fertilised egg. It is richly supplied with blood vessels and glands, which provide nourishment and support to the developing embryo. It also secretes hormones and other substances that are essential for the growth and development of the embryo. If there is no fertilisation, the corpus luteum simply regresses or degenerates. This leads to menstruation whereby the endometrial linining sheds.

173. *(b)* A is true and R is false. The juxtamedullary nephrons have a long Henle's loop that extends deep into the renal medulla, whereas the cortical nephrons have a shorter Henle's loop that only extends into the outer section of the renal medulla.

174. *(a)* Radial symmetry is a type of symmetry found in many animals, in which the body is arranged in a circular pattern around a central axis. This means that any plane passing through the centre of the animal divides it into two identical halves. Coelenterata (Cnidarians) and Ctenophora (Comb jellies) are two phyla that exhibit radial symmetry in their adult stages. Echinodermata (Starfish, sea urchins, etc.) exhibits bilateral symmetry in larval stage and radial symmetry in adult stage. Hemichordata (acorn worms) exhibits bilateral symmetry in their adult stages.

175. *(d)* A-III, B- I, C- IV, D- II

Fovea is the point of greatest visual activity or resolution. It is a small depression in the retina of the eye where visual acuity is highest. Iris is the visible coloured portion that regulates diameter of pupil, allowing more or less light to enter the eye. Blind spot is the point where optic nerve leaves the eyeball and photoreceptor cells are absent. Sclera is the external layer of eye formed of dense connective tissue. The sclera is the tough, white outer layer of the eye that protects and supports the eyeball, and provides attachment points for the muscles that move the eye.

176. *(a)* Statement 1 is correct while 2, 3, and 4 are incorrect. These can be corrected as.

Algal blooms are caused by an excess of nutrients in the water, particularly nitrogen and phosphorus. Algal blooms can cause a variety of problems for fish, including depleting oxygen levels in the water and producing toxins that can harm or kill fish. Eutrophication is the process by which a body of water becomes enriched with nutrients, often due to runoff from land. This enrichment can come from a variety of sources, including sewage and waste water, agricultural runoff, etc.

177. *(c)* Statements A, C, and D are correct whereas B is incorrect.The first menstrual cycle in human females typically begins at puberty, but menopause marks the end of menstrual cycles, not the beginning of it.

178. *(c)* Statement I is incorrect but statement II is true. DNA is a negatively charged molecule whereas proteins are positively charged molecules.In prokaryotes, the DNA is not enclosed within a membrane-bound nucleus, and it is present in the nucleoid region.

179. *(a)* The correct group/set of Australian Marsupials exhibiting adaptive radiation is; Numbat, spotted cuscus, flying phalanger. Adaptive radiation is the evolutionary divergence of members of a single group into a variety of forms, each adapted to a specific environment or mode of life. The three marsupials mentioned in option (1) have diversified into different forms that are adapted to different habitats and feeding strategies, such as arboreal (tree-dwelling) and terrestrial (ground-dwelling). The other options do not contain marsupials that have undergone significant adaptive radiation.

180. *(b)* A- III, B- II, C- IV, D- I

Taenia is a parasitic flatworm that lacks a specialised excretory system. Instead, it has specialised flame cells which help in excretion. *Paramoecium* is a unicellular organism with a contractile vacuole that helps in the regulation of water content by collecting and expelling excess water from the body. *Periplaneta* is a species of cockroach that has specialised uric acid-secreting glands called urecose glands, which help in the excretion of nitrogenous waste products. *Pheretima* or earthworms possess nephridia as excretory structures.

181. *(c)* A is false, but R is true.Amniocentesis for sex determination is not a strategy of Reproductive and Child Health Care Programme. In fact, sex determination tests are banned in many countries, including India, due to the increasing menace of female foeticide. Amniocentesis is a medical procedure that involves the collection of a small amount of amniotic fluid from the amniotic sac surrounding the developing foetus. This procedure is primarily used to diagnose genetic disorders and chromosomal abnormalities in the foetus and not for determining the sex of the foetus.

182. *(c)* Probe is not a cloning vector.A cloning vector is a DNA molecule used to carry foreign genetic material into a host cell for the purpose of replicating or expressing it. A probe, on the other hand, is a short, single-stranded DNA or RNA fragment that is used to detect the presence of a specific target sequence in a complex mixture of nucleic acids, such as in a Southern or Northern blot.

183. *(b)* Statement I is correct but statement II is incorrect.

An electrostatic precipitator is an efficient dust collection device that is widely used in thermal power plants and other industries to remove particulate matter from exhaust gas streams. However, in a thermal power plant, it does not help to remove ionising radiations. Ionising radiation is a type of radiation that has enough energy to ionise atoms or molecules, causing them to become charged.

184. *(d)* A- III, B- I, C- IV, D- II

The P-wave in an electrocardiogram (ECG) represents the depolarisation (contraction) of the atria.The Q-wave is the first negative deflection in the QRS complex and marks the beginning of ventricular depolarisation and the onset of systole.The QRS complex in an ECG represents the depolarisation of the ventricles, which leads to the contraction of the ventricles.The T-wave in an ECG represents the repolarisation (relaxation) of the ventricles after contraction.

185. *(c)* Vital capacity (VC) is the maximum amount of air that a person can exhale after taking a deep breath. It is a measure of lung function and is commonly used to assess respiratory health.It is calculated by summing tidal volume, inspiratory reserve volume, and expiratory reserve volume. VC = TV + IRV + ERV.

186. *(c)* Statement I is incorrect but statement II is correct.

During the G_0-phase of cell cycle, cell is metabolically active. The G_0-phase of the cell cycle is a period of quiescence, where cells are not actively dividing. Some cells in the G_0-phase, such as nerve cells, muscle cells, and some immune cells, are highly metabolically active and perform their specialised functions even though they are not actively dividing.

187. *(d)* Limbic system and hypothalamus helps in the regulation of sexual behavior, expression of excitement, pleasure, rage and fear. The limbic system is a complex network of brain structures that are involved in the processing and regulation of emotions, motivation, and memory. The hypothalamus is a small but important structure located in the brain that is involved in the regulation of many bodily functions, including temperature control, hunger and thirst, and the release of hormones.

188. *(b)* Among the given options, B, C and D are correct. In cockroach, excretion is brought about by urecose gland, nephrocytes and fat body. The urecose gland, also known as the urinary bladder, is involved in the storage and elimination of urine in cockroaches. The nephrocytes, which are specialised cells found in the fat body, are also involved in the removal of nitrogenous waste products. The phallic gland and collateral glands are not involved in excretion in cockroaches. The phallic gland is a gland associated with the male reproductive system, while the collateral glands are involved in the production of accessory gland secretions that are important for sperm storage and transfer.

189. *(a)* Among given options, B, C, and D are correct. Statements A and E are incorrect and can be corrected as

Excessive loss of body fluid, such as through sweating, vomiting, or diarrhoea, can lead to dehydration and an increase in the concentration of solutes in the blood. This can activate the osmoreceptors and stimulate the release of ADH to help restore the body's fluid balance.ADH is antidiuretic hormone which prevents diuresis but it also causes constriction of blood vessels thus increasing the glomerular filtrate rate (GFR).

190. *(c)* Among given statements, option C does not represent the advantage of inbreeding.When inbreeding occurs, the genetic diversity within the population decreases, which can increase the expression of harmful recessive traits and decrease the overall productivity of the population. Hence, it is a drawback of inbreeding.

191. *(a)* Among the given statements, B and C are correct while A and D are incorrect can be corrected as

- Skeletal muscle fibers are bundled together into fascicles by a connective tissue layer called the perimysium. The perimysium is a collagenous connective tissue layer that surrounds each fascicle and helps to protect and support the muscle fibers within it.
- The sarcomere is the basic functional unit of muscle contraction. It is a highly organised structure that is composed of actin and myosin filaments, as well as other proteins that regulate muscle contraction. The M-line is a region in the center of the sarcomere that anchors the myosin filaments and helps to maintain the structural integrity of the sarcomere during muscle contraction.

192. *(d)* A-II, B- I, C- III, D- IV

- Logistic growth occurs when a population grows rapidly initially, but then levels off due to limited resources such as food, waterorspace. The growth rate slows down and eventually reaches a carrying capacity where the population size stabilises.
- Exponential growth occurs when a population grows at a constant rate, without any constraints on resources. This type of growth is

rarely observed in natural populations.

- An expanding age pyramid occurs when a population has a large proportion of young individuals who are not yet reproductive, and a smaller proportion of older individuals who have already passed their reproductive years.
- A stable age pyramid occurs when a population has roughly equal proportions of individuals in each age group, with similar numbers of pre-reproductive, reproductive, and post-reproductive individuals.

193. *(b)* The coding strand is the DNA strand that has the same nucleotide sequence as the mRNA transcript, except that thymine (T) is replaced with uracil (U) in the RNA molecule.

Hence the sequence of given coding strand will be as follows

5'ATCGATCGATCGATCGATCGATCGAT CG 3'

194. *(b)* Among given statements, B and C are correct whereas A, D and E are incorrect.These statements can be corrected as

- Basophils are the least abundant cells of total WBCs.
- Basophils have a bilobed nucleus, which gives them an S-shaped or irregularly shaped appearance.
- Basophils are granulocytes, not agranulocytes.

195. *(b)* A-II, B- I, C- IV, D- III

- Mast cells are a type of white blood cell that are found in connective tissue throughout the body. They play a role in the immune response and the development of allergies. Areolar tissue is a loose connective tissue that contains a variety of cell types, including mast cells.
- The bronchioles are small tubes that branch off from the bronchi in the lungs. The inner surface of the bronchioles is lined with a layer of ciliated epithelial cells that have hair-like projections called cilia.
- Blood is a type of specialised connective tissue that circulates throughout the body, delivering oxygen and nutrients to cells and tissues and removing waste products.
- The tubular part of the nephron is lined with a layer of cuboidal epithelial cells. These cells play a role in reabsorbing water and other substances from the filtrate that is produced by the glomerulus.

196. *(a)* The anal style is a modified structure on the last abdominal segment of male cockroaches that is absent in females. The anal style is used by male cockroaches during mating to help grip the female during copulation.

197. *(b)* Statements C and D represents the activities which are not under the control of thyroid hormone.The normal rhythm of sleep-wake cycles is under the control of the pineal gland.It is responsible for regulating the production and secretion of the hormone melatonin, which is involved in the timing of the sleep-wake cycle. On the other hand, the development and regulation of the immune system is under the control of the endocrine system, particularly the thymus gland.

198. *(a)* Among the given options, B and D are correct whereas A, C, and E are incorrect. These can be corrected as

- Tetrad formation is seen during pachytene stage of prophase-I, of meiosis.
- Terminalisation of chiasmata occur during diakinesis stage of prophase-I, of meiosis.
- Crossing over takes place between non-sister chromatids of homologous chromosomes.

199. *(a)* The unique mammalian characters are hair, pinna and mammary glands. Hair provides insulation and protection, pinna (external ear) is used for hearing, and mammary glands produce milk for nourishing young. These features are exclusive to mammals and play important roles in their survival and reproduction.

200. *(a)* Among given options, statements B and C are correct whereas A, D, and E are incorrect.These can be corrected as

- Chordates possess a hollow, dorsal nerve cord. The nerve cord is not double but is a single tube-like structure located dorsal to the notochord.
- Chordates possess a ventral heart. The heart is located on the ventral side of the body, not the dorsal side.
- Chordates are triploblastic coelomate animals. They possess three germ layers (endoderm, mesoderm, and ectoderm) and a true coelom, which is a body cavity completely lined by mesoderm.